

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: DskEth-Rev-Ce.sil etc.

Creation date: 23-Sep-79 20:54:21

For: Spruce

49 total sheets = 48 pages, 1 copy.

Problems encountered:

Font HELVETICA24B substituted for font HELVE24.

Character code 57b not found in font Gates32.

Character w/ code 105b, font Helvetica7I would lie off the page.

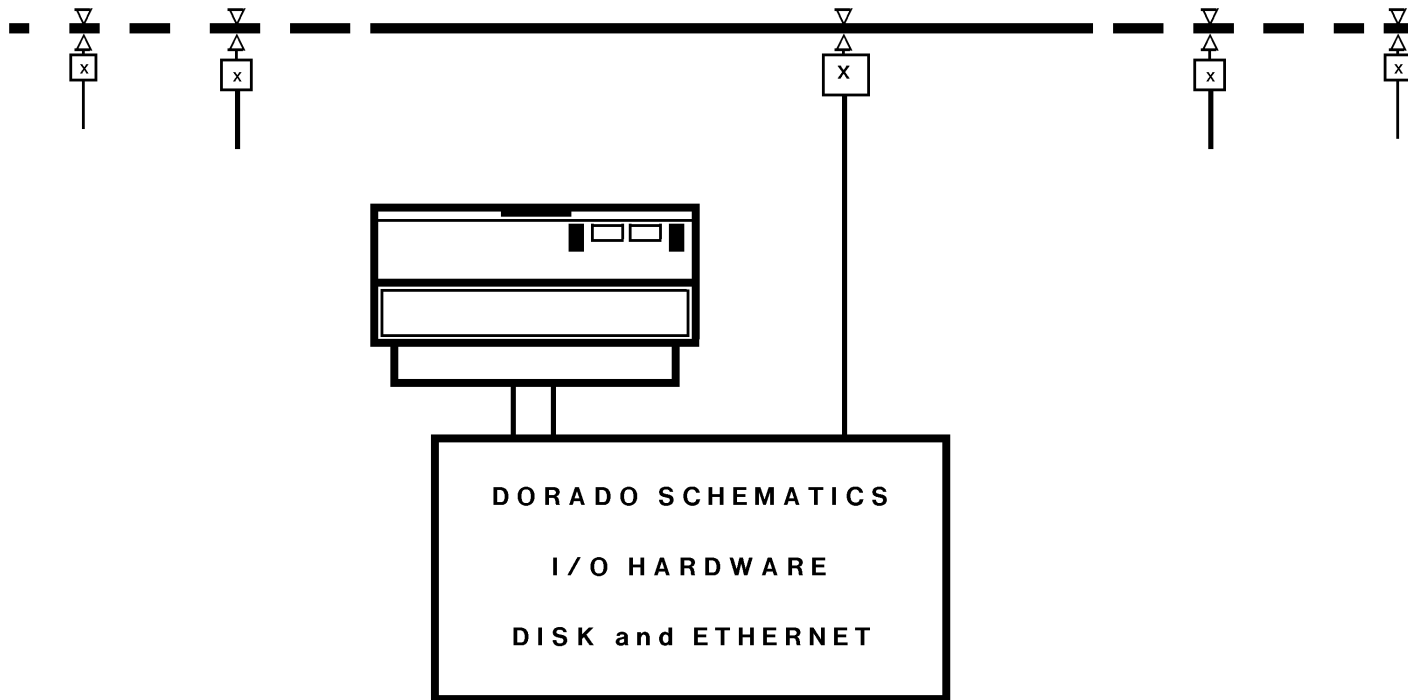
Character code 117b not found in font Gates32.

Character code 157b not found in font Gates32.

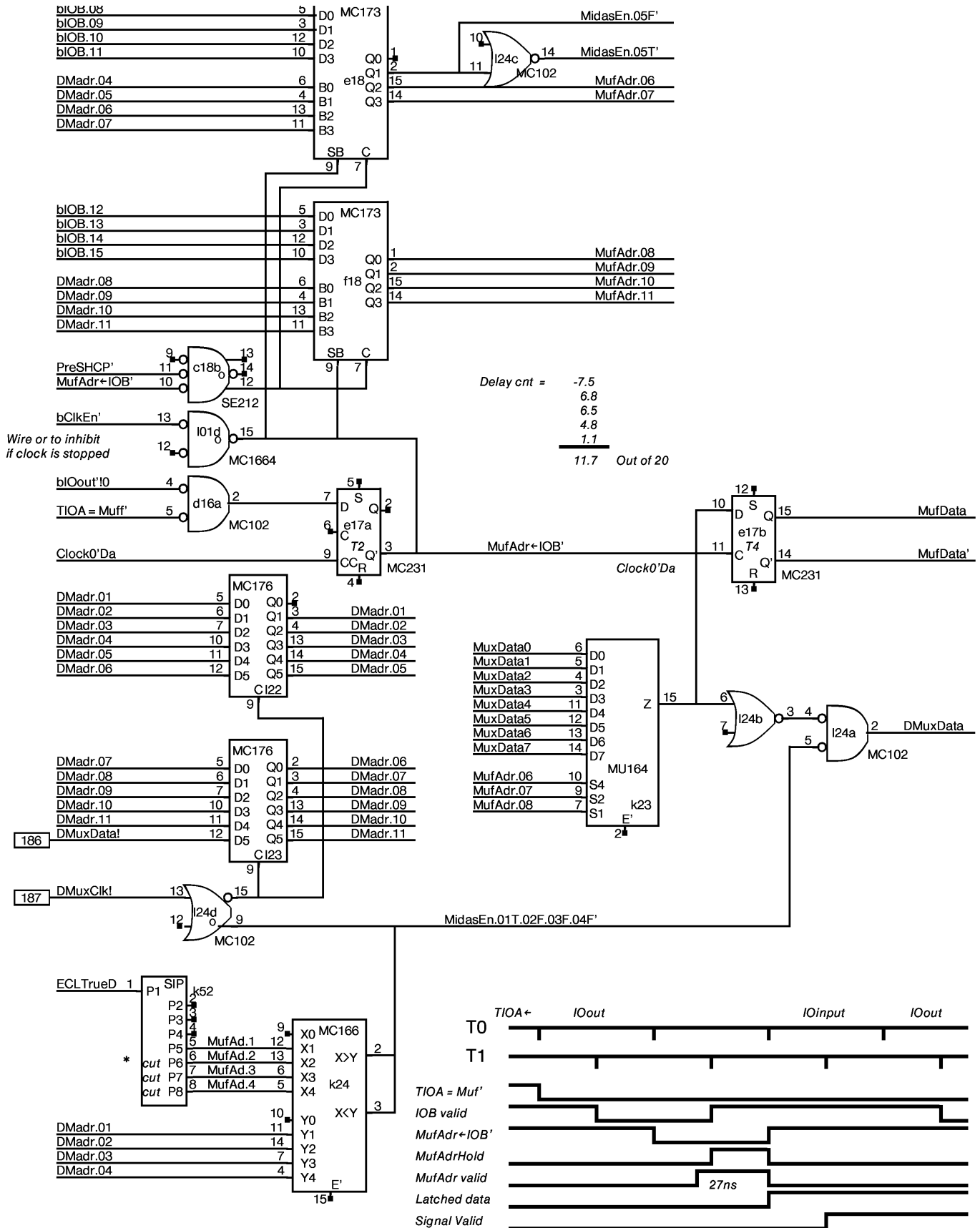
Character code 117b not found in font Gates32.

Character code 157b not found in font Gates32.

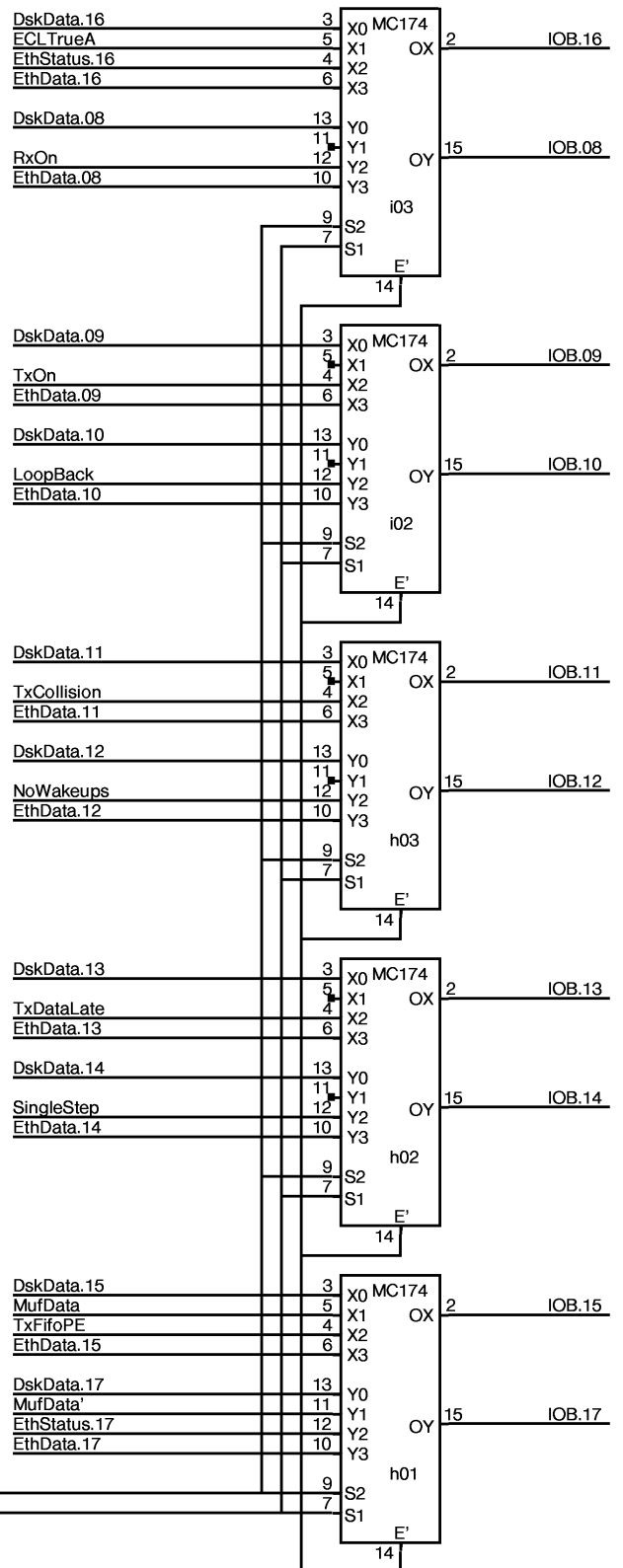
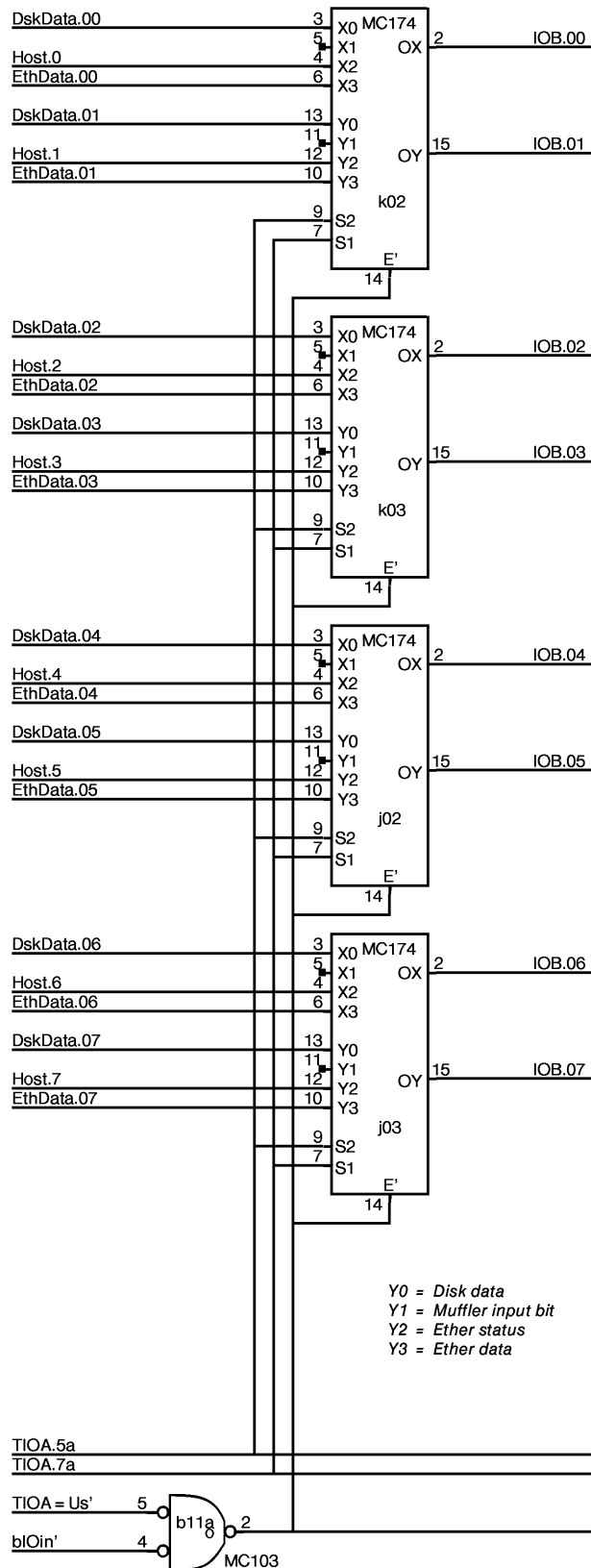
Character code 117b not found in font Gates32.

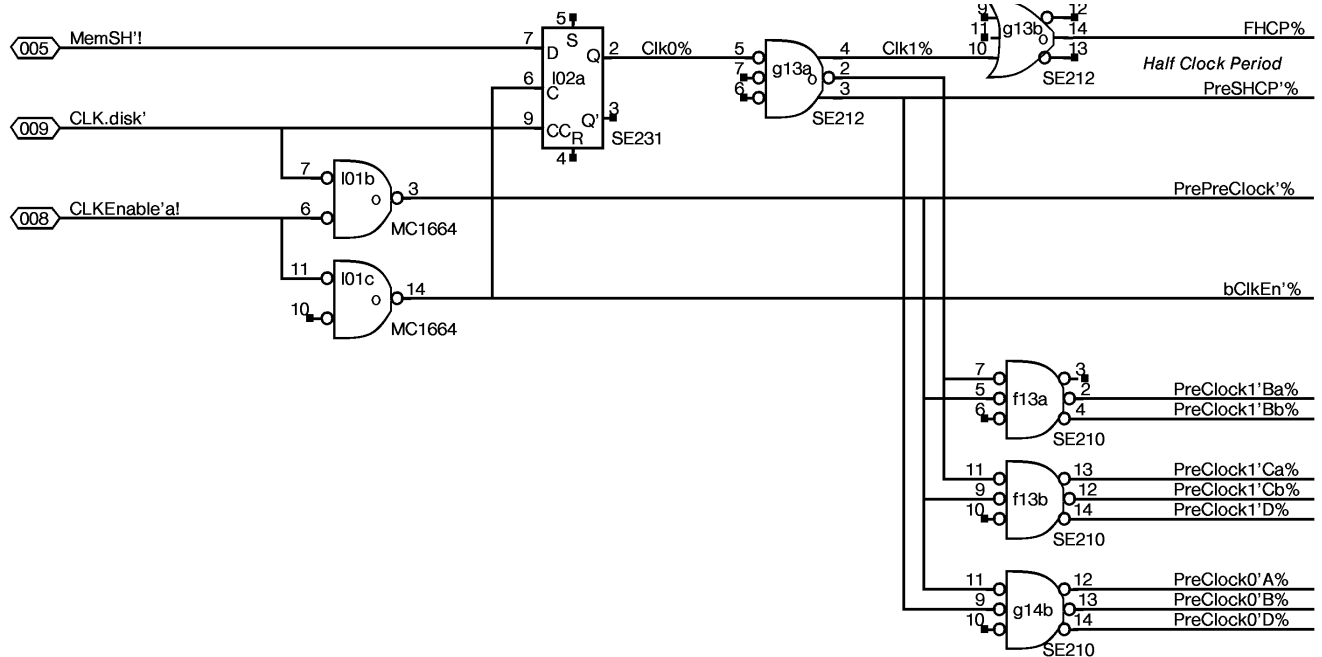


1) Drawings of common logic	01
Midas Muffler Control	01
IOA and IOB	02
Clocks and Temp sense	04
Layout	05
Configuration	06
2) Drawings for TriconD disk Controller	07
State Control Register	08
Format Ram, Counter and Proms	09
Tag Register	10
Disk Drive Control	11
FIFO	13
Error Correction Shift Register	15
Task Wake-Up and IOB parity check	16
Mufflers	18
Clocks	19
I/O pins and Termination	20
Timing Diagram	21
Cable Assembly Drawings	22
3) Drawings for Ethernet Controller	23
Receiver	24
Transmitter	29
Test Logic	34
Clocks	35
Next Bus and IOattention	36
Mufflers	37
Cable I/O and termination	38
Timing Diagrams	40

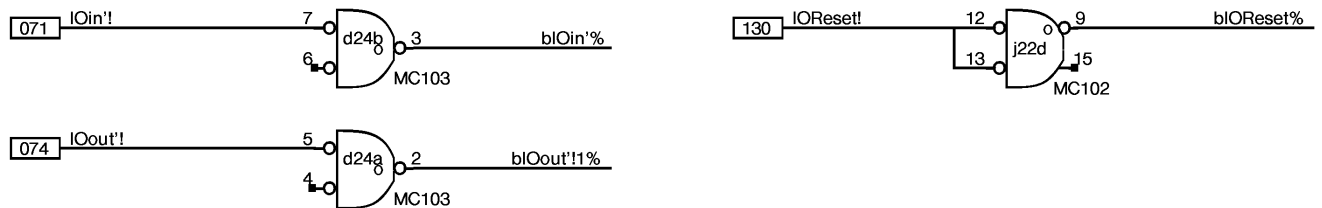


* Standard muffler addresses are 2000-2177
See DskEth06.sil for information
on how to set other addresses

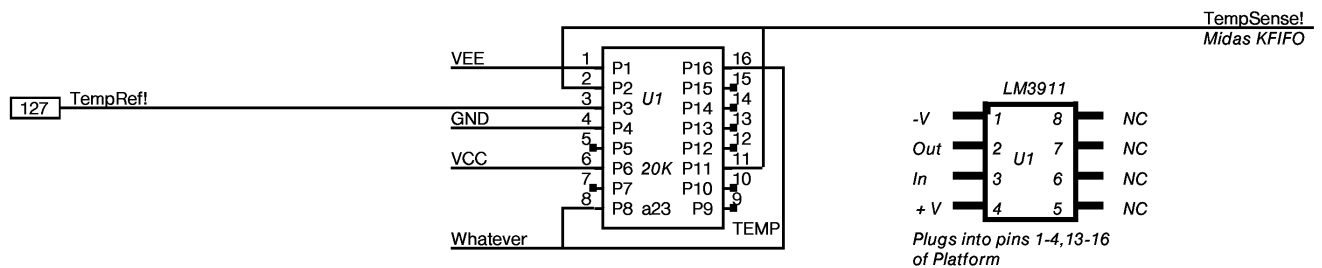




Common Clock circuitry



Common IO circuitry

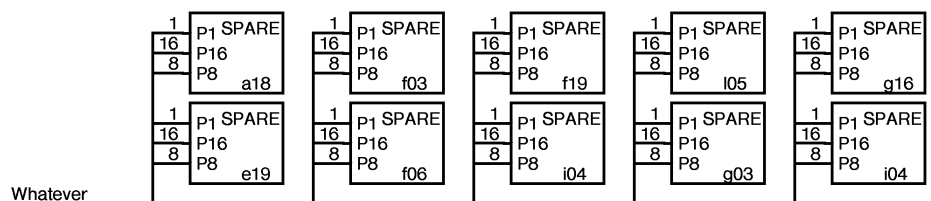


Temperature Sensor

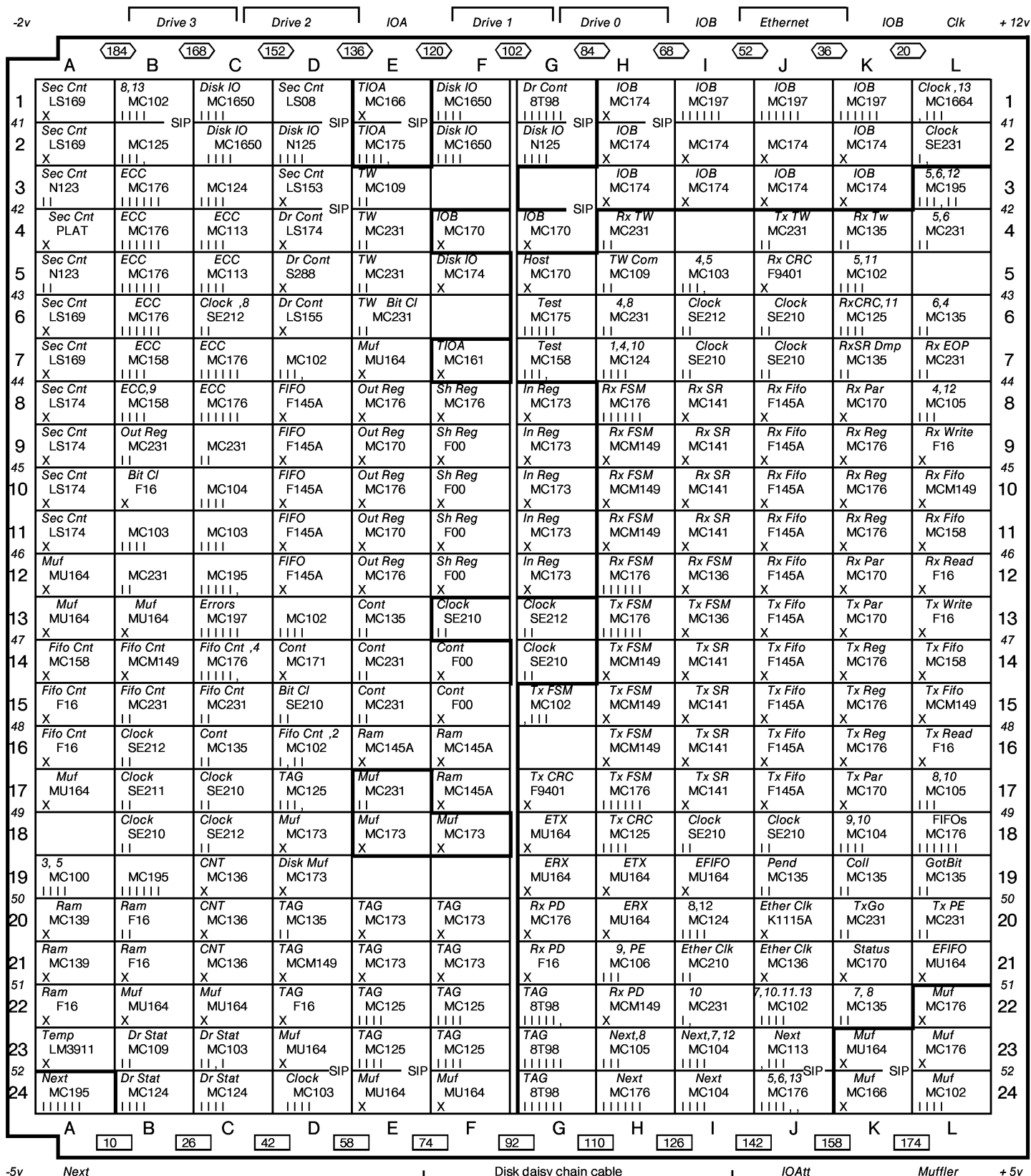
Power connections are as follows:

Stitch-Weld: 1 & 16 are GND
8 is -5

Multiwire: 16 is GND
8 is -5
except in locations
a18, g03, and g16
which are uncommitted



Spare Socket locations for Multiwire



31 chips common to Disk & Ether
 137 chips specific to Disk
 111 chips specific to Ether
 9 spare chip positions
 288 total chip positions

* Standard addresses are 2000-2177.

	1800-1777	cat			
*	2000-2177		cut	cut	cut

* Standard tasks are 6 & 7.

	4 & 5	cut
*	6 & 7	cut cut

Name	Type	#	Location		
DskEth					
Disk					
DiskRead	SG139	2	a20	a21	
DiskTag	MC149	1	d21		
DiskUnits	S288	1	d05		
DiskFifo	MC149	1	b14		
Ether					
EtherPD	MC149	1	h22		
EtherRcvr	MC149	3	h09	h10	h11
EtherFifo	MC149	2	l10	l15	
EtherXmtr	MC149	3	h14	h15	h16

"DoradoProms DskEth" gets all proms for the DskEth board.
"DoradoProms Disk" gets all proms for the Trident half of the board.
"DoradoProms Ether" gets all proms for the Ethernet half of the board.

* Standard addresses are 10-17.

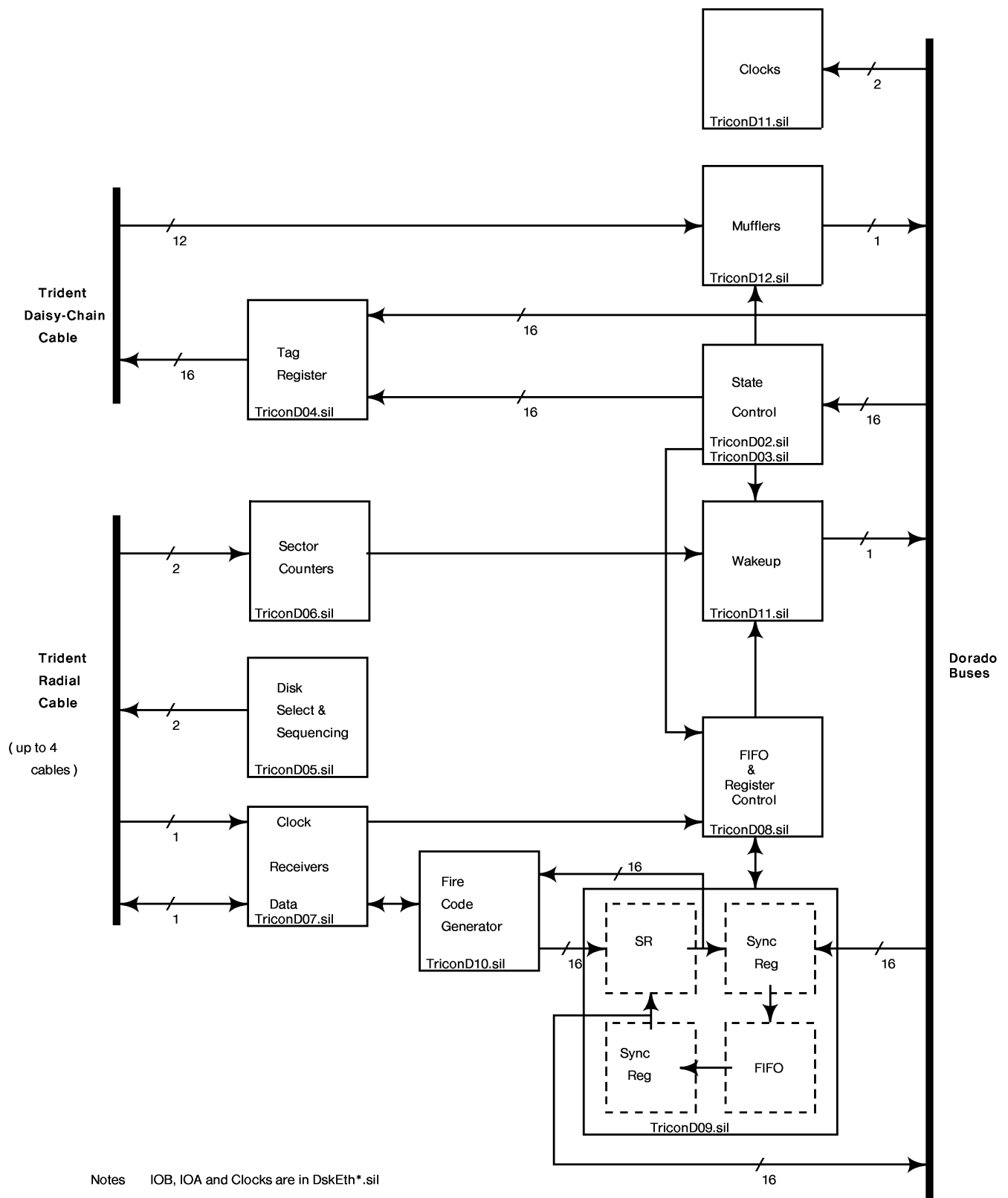
	000-007	cut	cut	cut	cut	cut
*	010-017	cut	cut	cut	cut	

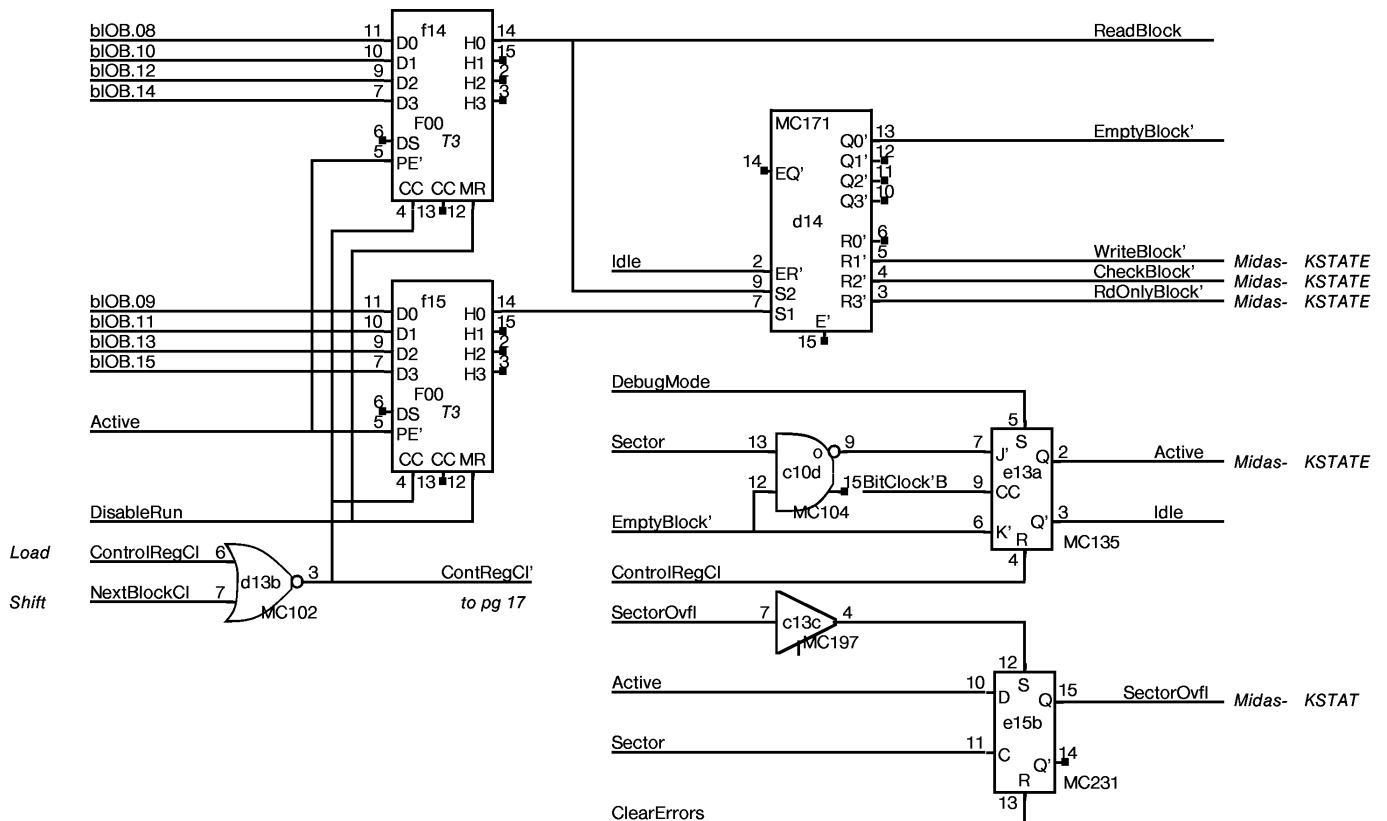
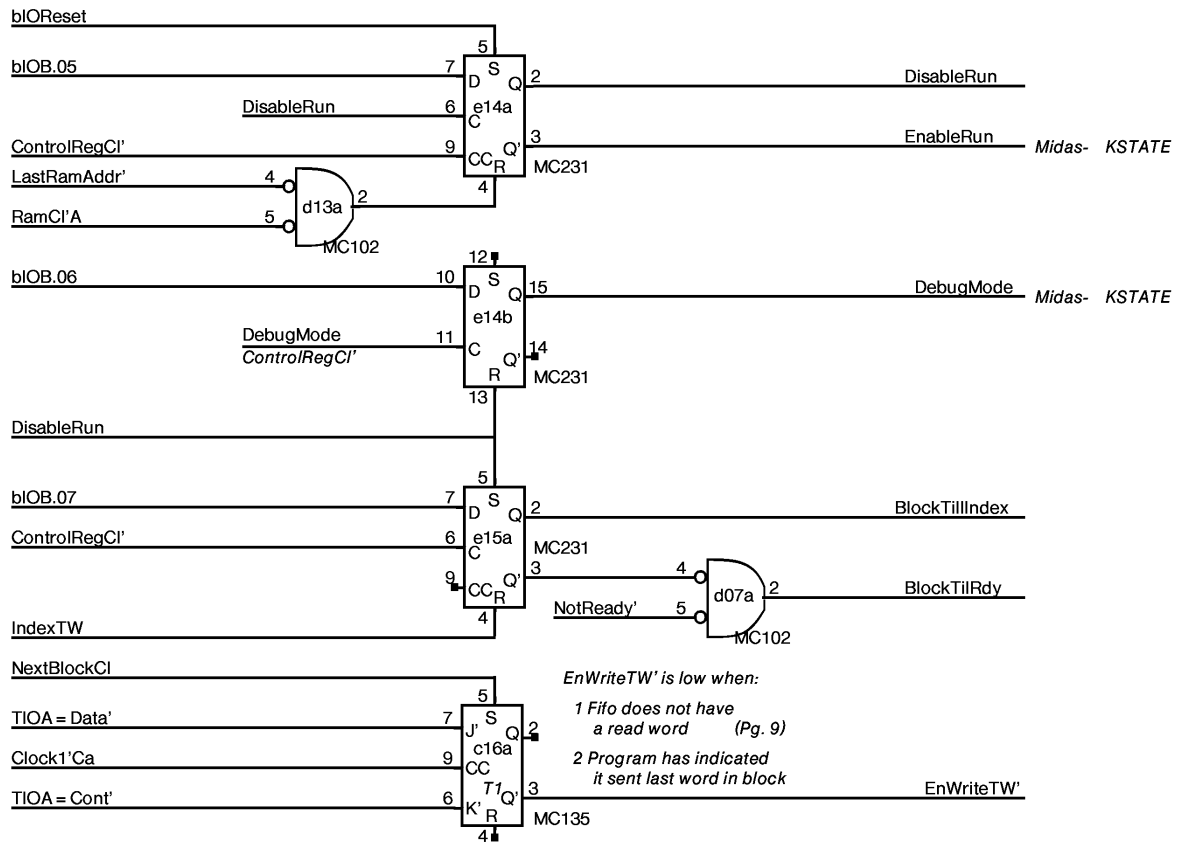
Name	Type	#	Location		
DskEth					
Disk					
DiskRead	SG139	2	a20	a21	
DiskTag	MC149	1	d21		
DiskUnits	S288	1	d05		
DiskFifo	MC149	1	b14		
Ether					
EtherPD	MC149	1	h22		
EtherRcvr	MC149	3	h09	h10	h11
EtherFifo	MC149	2	l10	l15	
EtherXmtr	MC149	3	h14	h15	h16

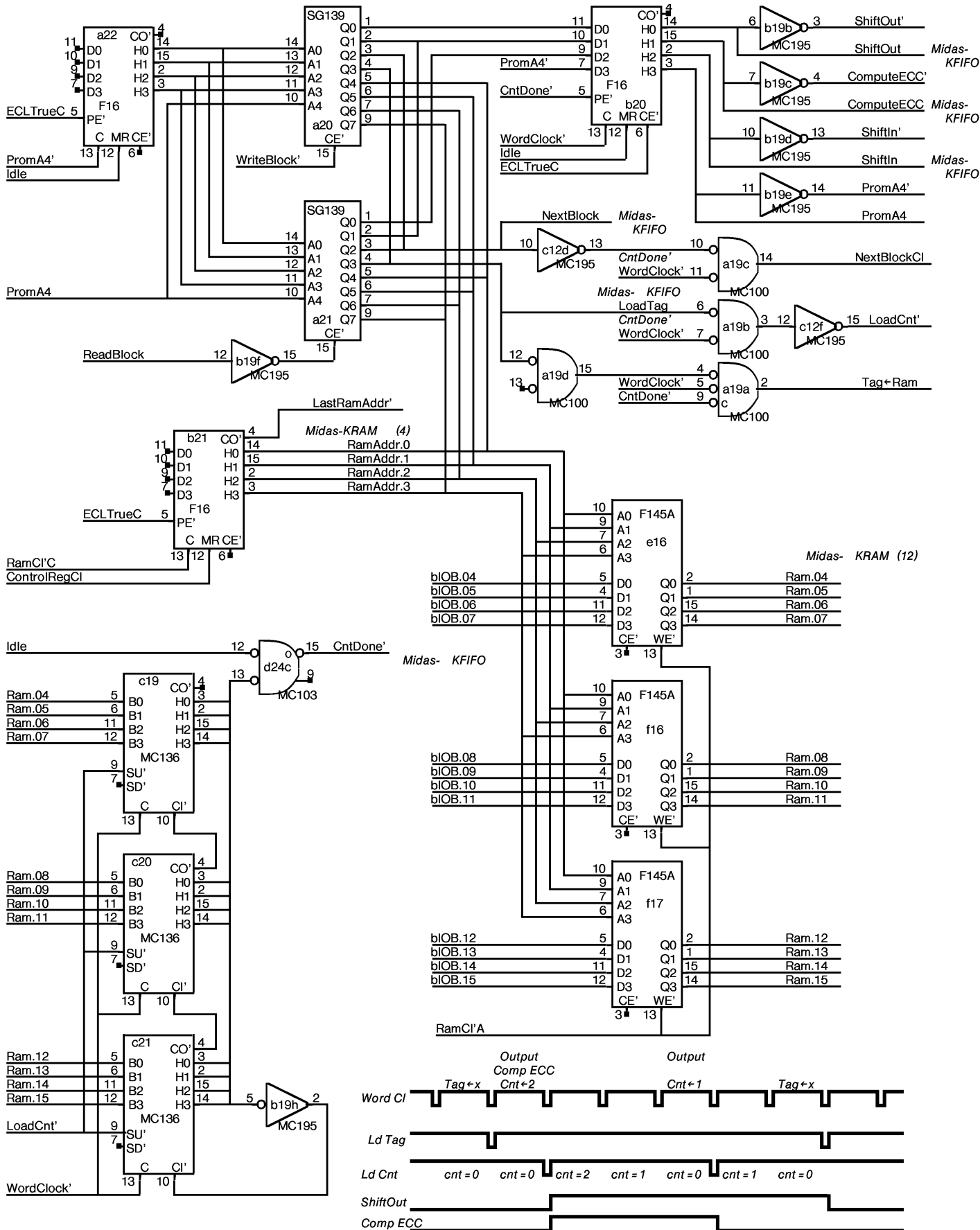
All resistors 27K
All capacitors 330pf

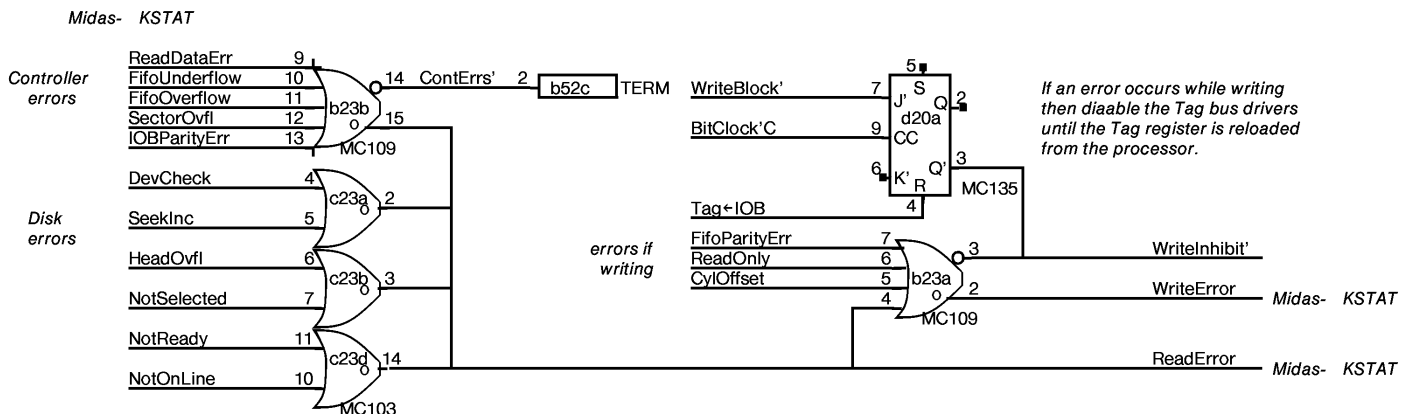
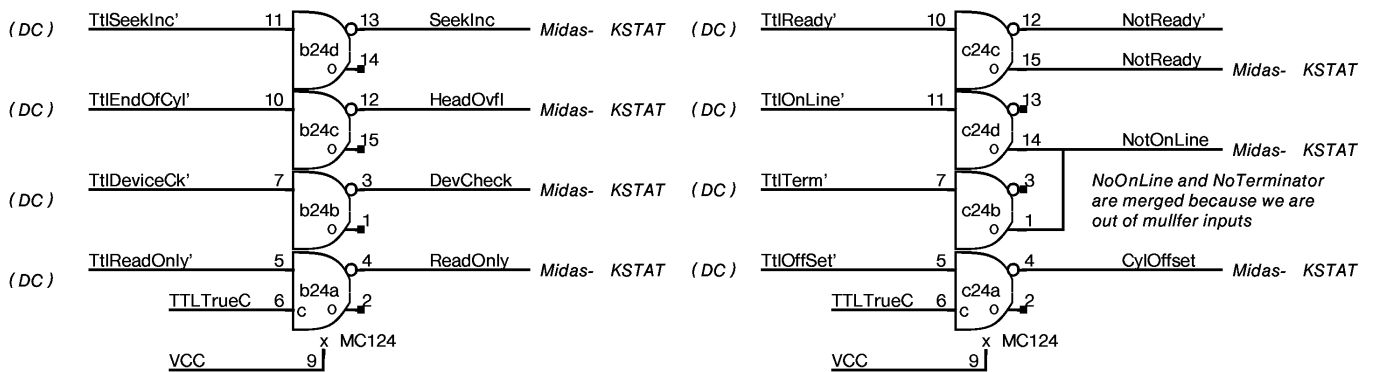
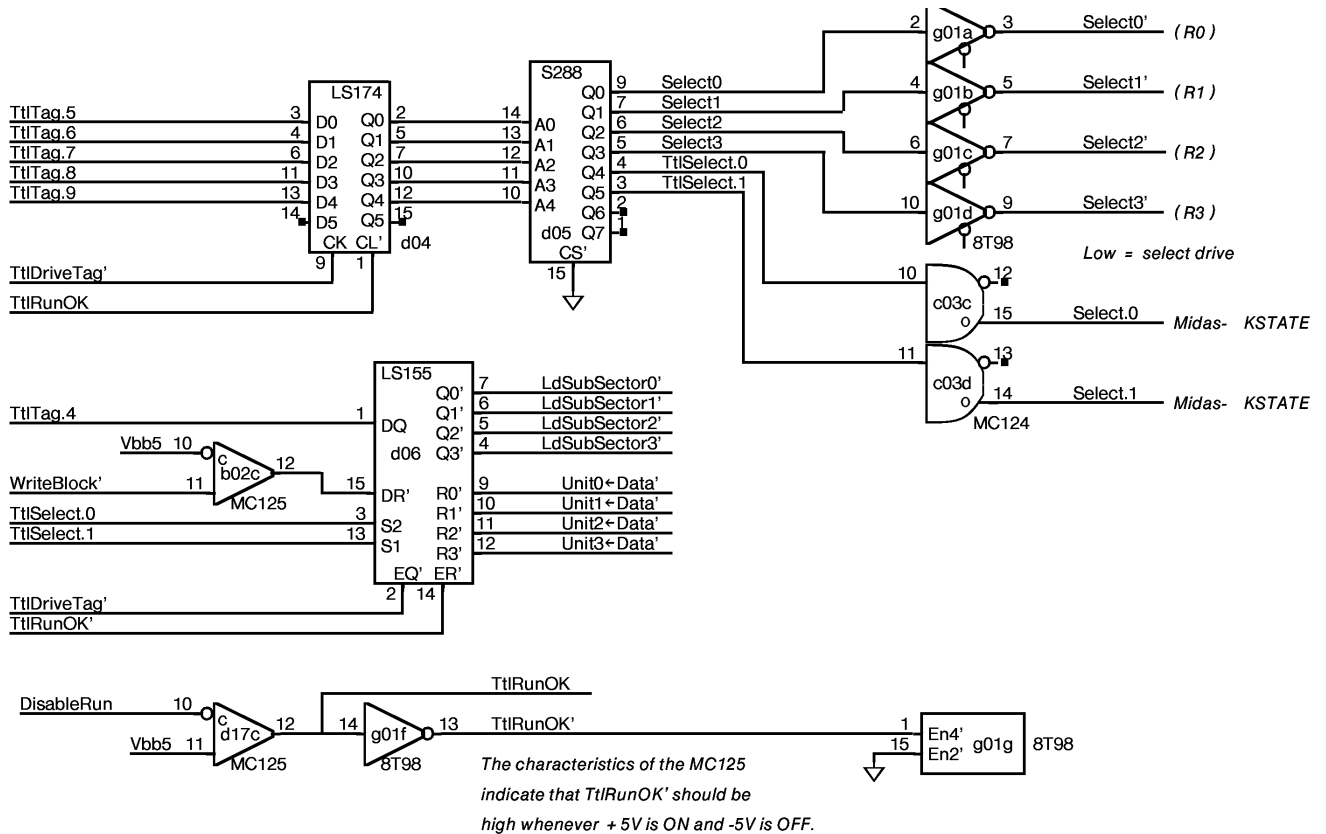
M
23.530 Mhz
xtal osc.

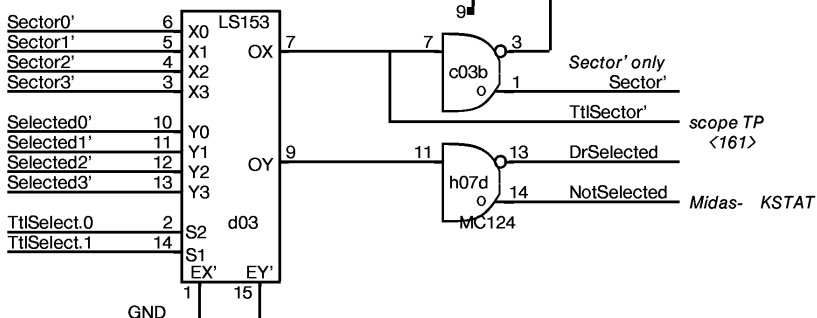
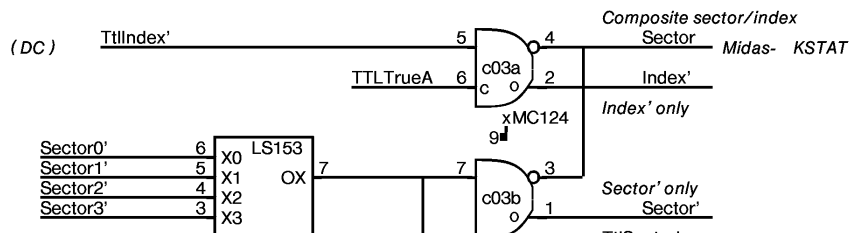
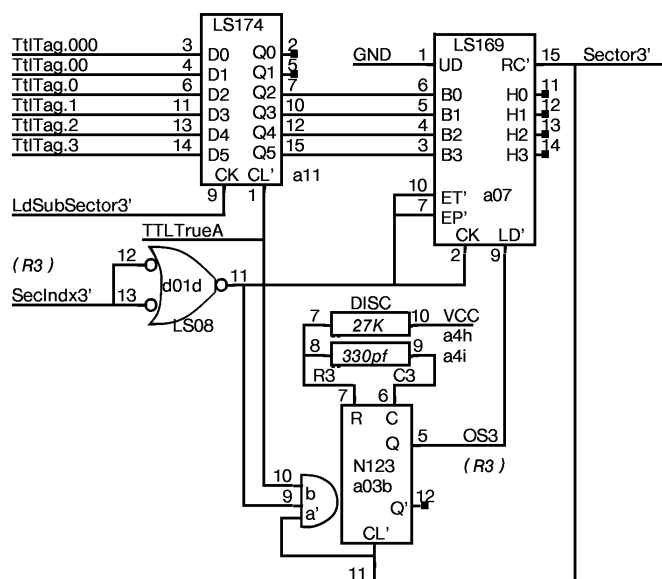
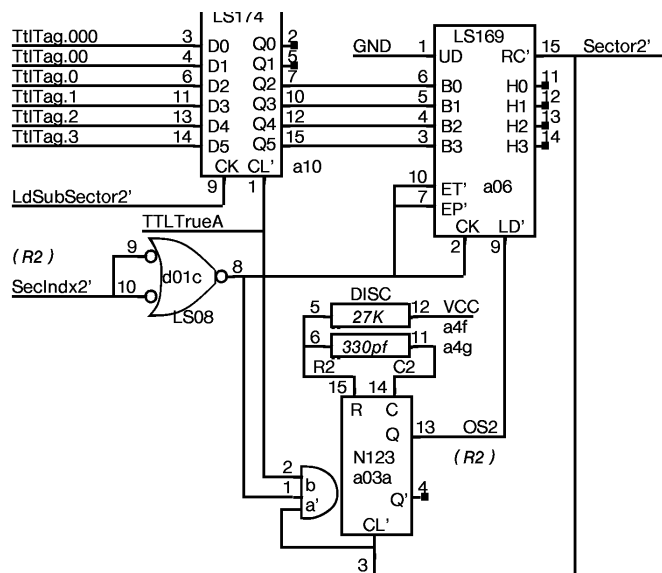
Logo is near pin 1.
This is a 14 pin pkg.
Board pins 8 & 9 aren't used.
Remove bypass cap above pin 1



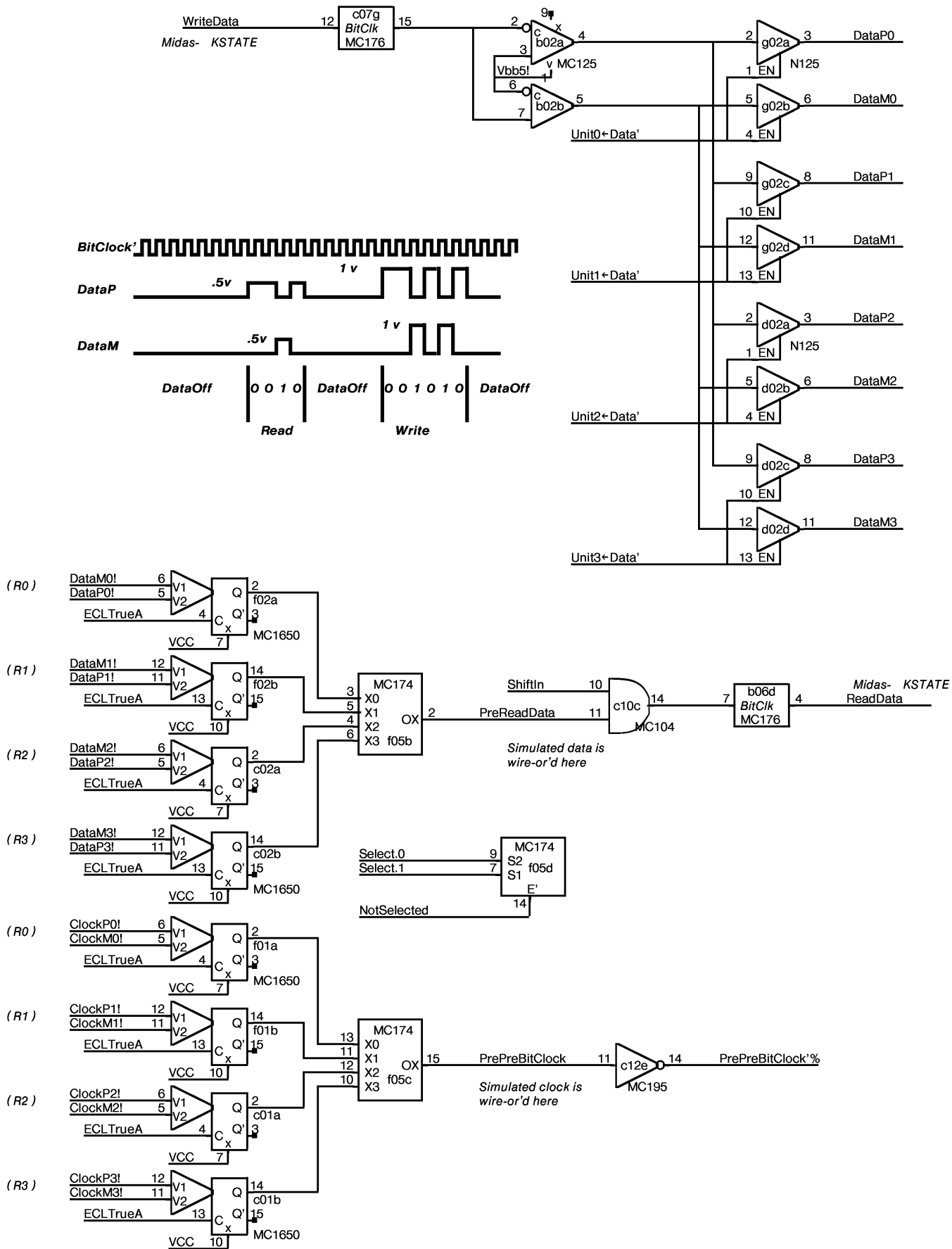






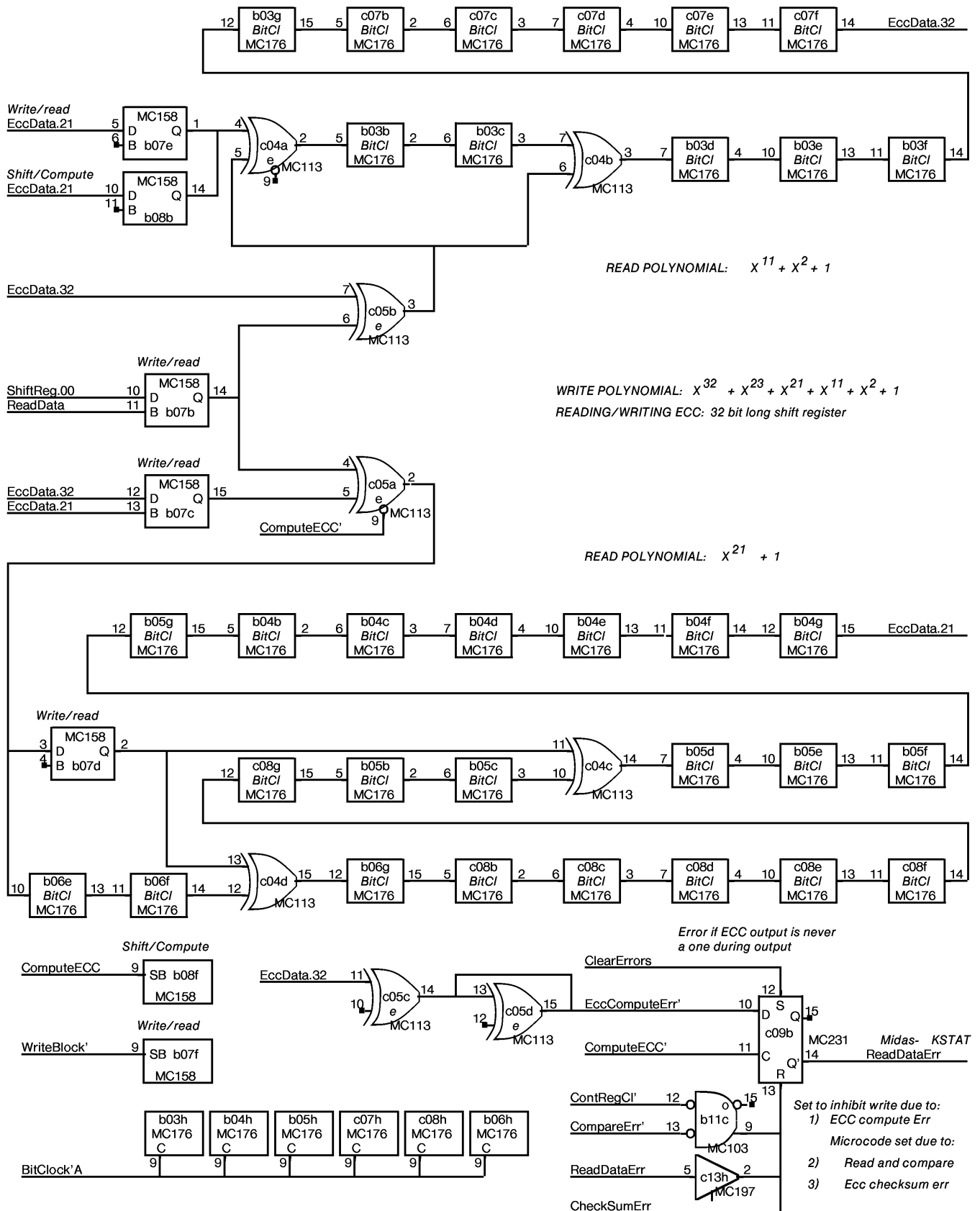


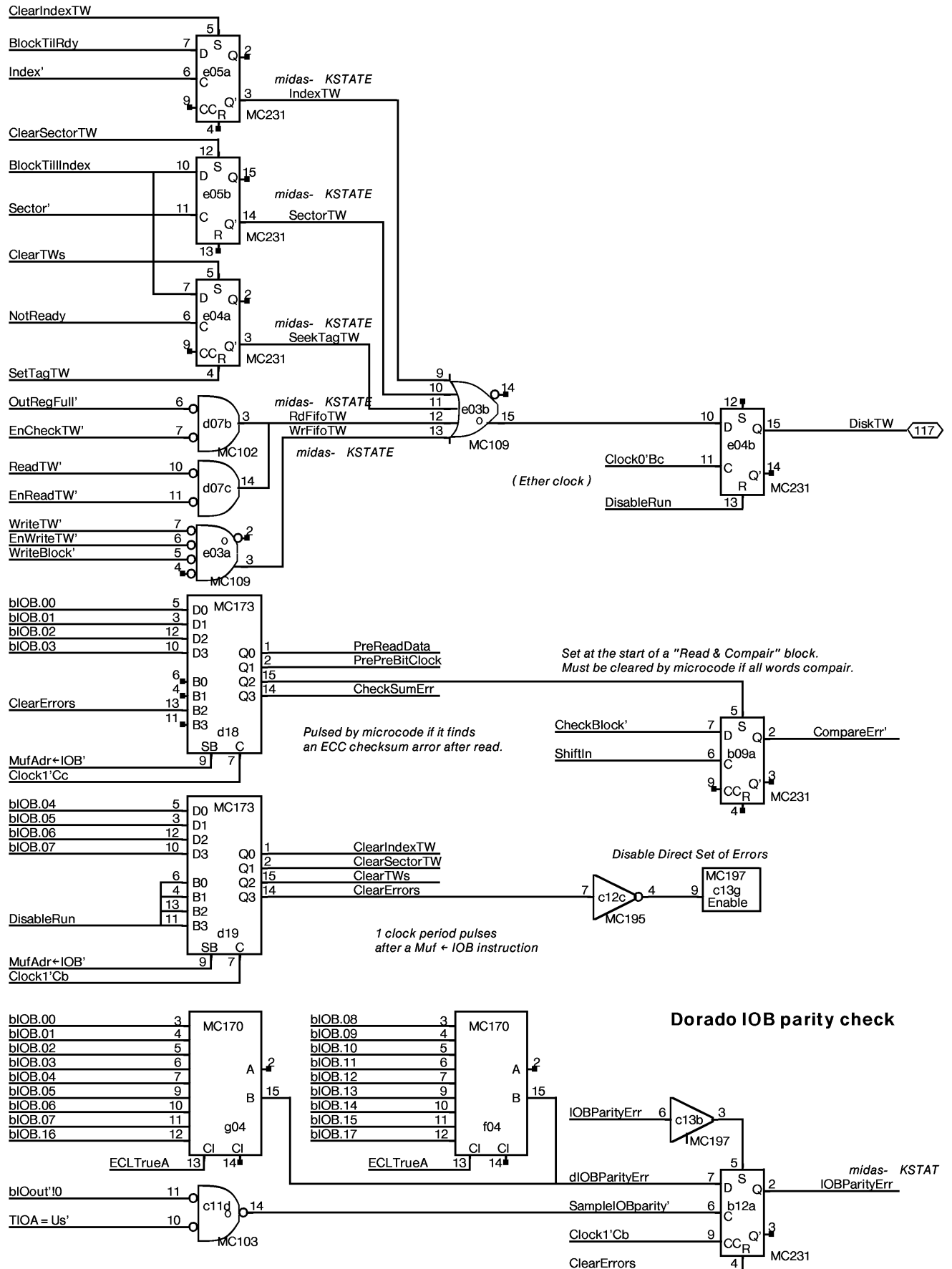
X6A	X6B
4 - 11	2 - 13
5 - 10	3 - 12
	4 - 11
	5 - 10
	6 - 09



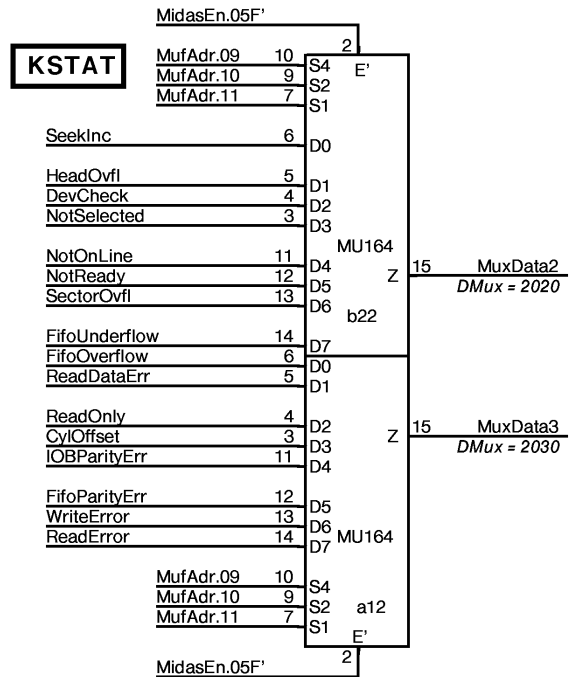
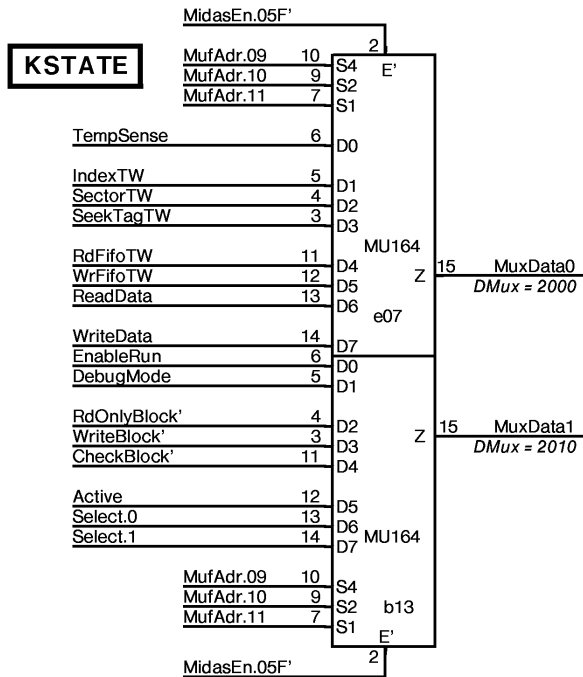


POLYNOMIAL DIVIDER FOR FIRE CODE GENERATION





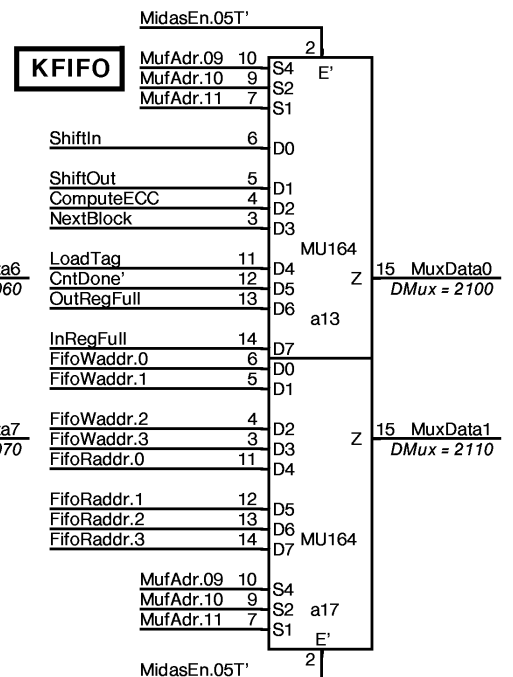
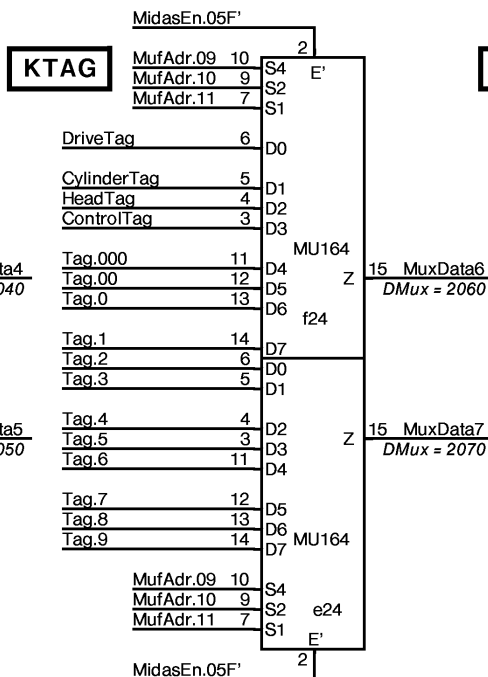
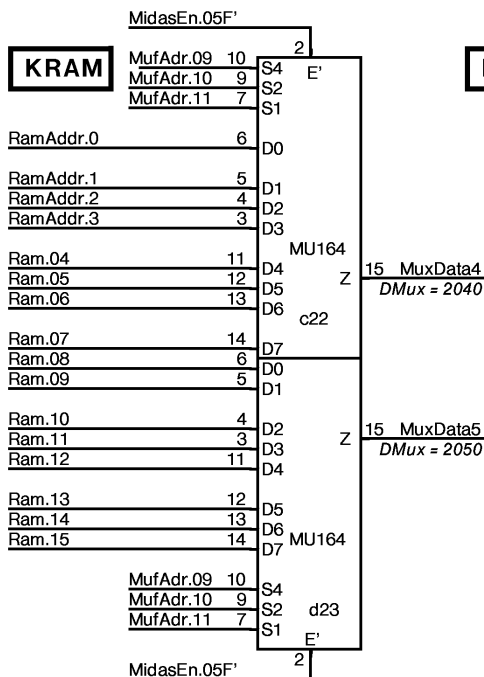
Dorado IOB parity check

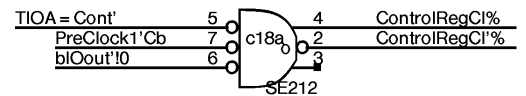
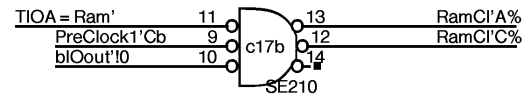
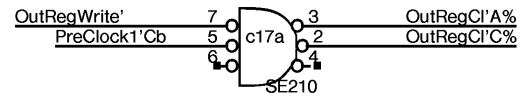
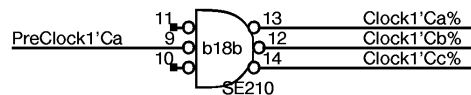
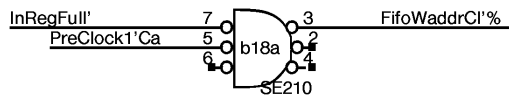
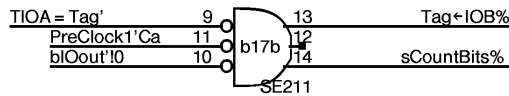
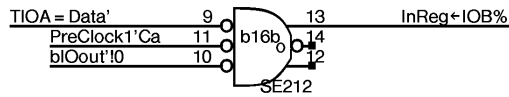
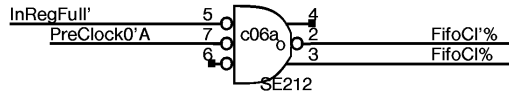
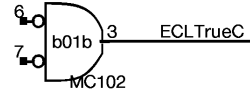
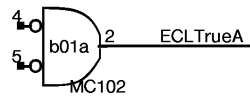


Muffler addresses are:

Value listed for Program input
Value plus 2000 for Midas input

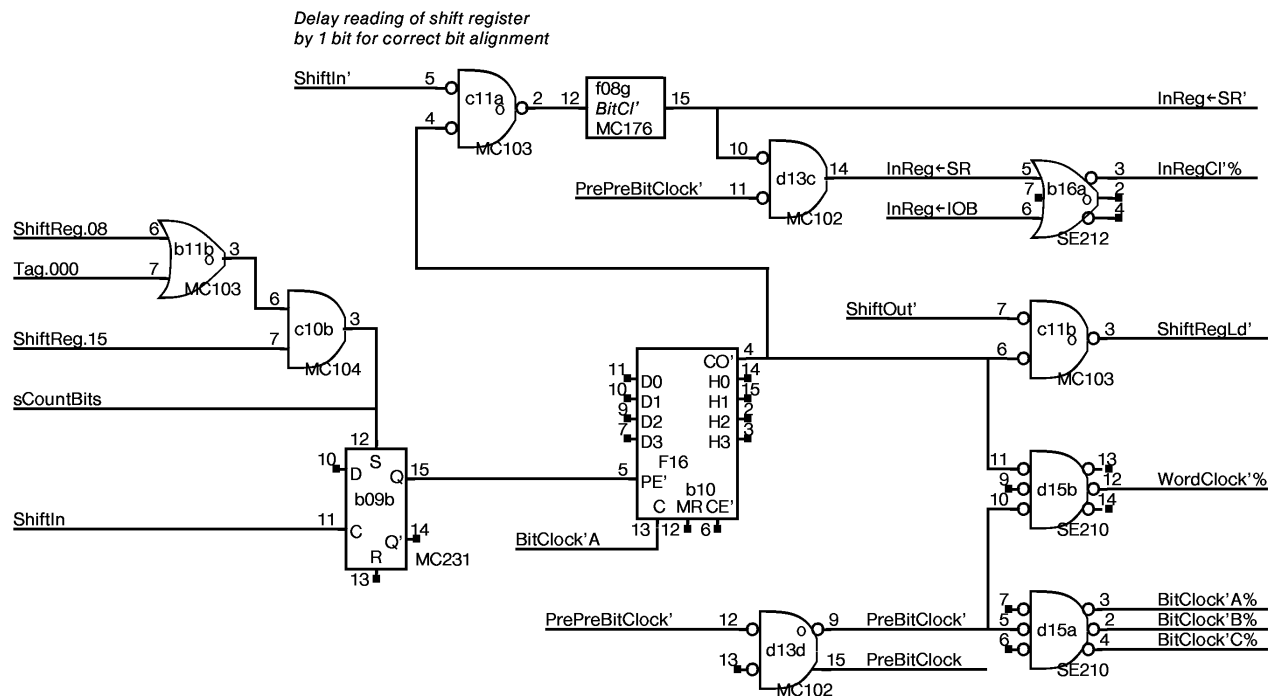
Values from 120 to 177 are
used by the Ethernet



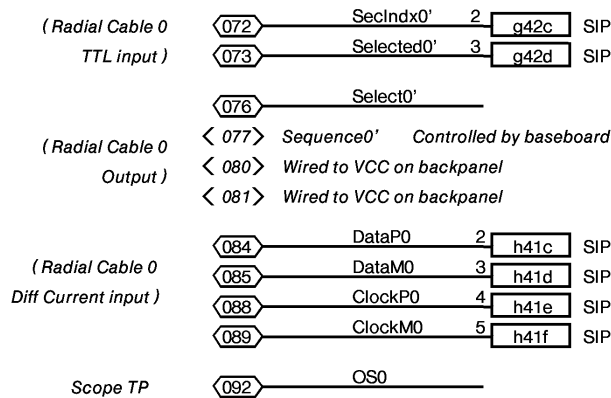


System Clocks

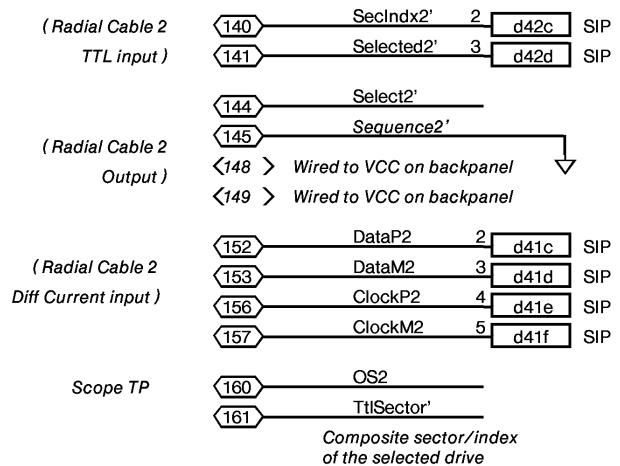
Disk Clocks



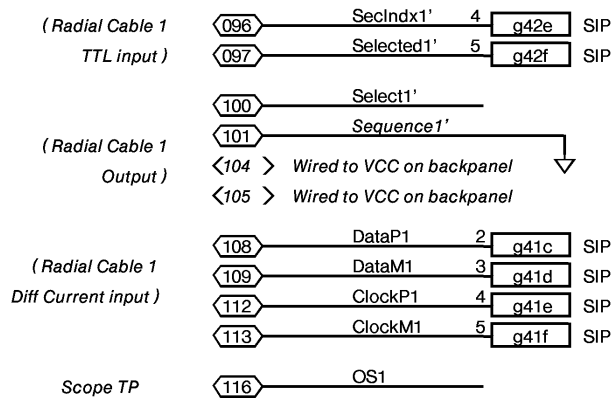
Radial Cable for Drive 0



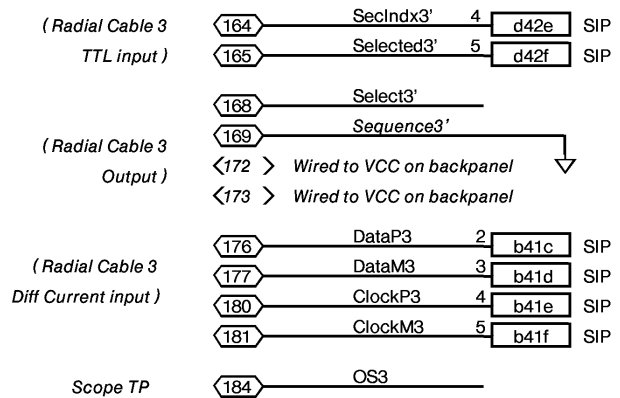
Radial Cable for Drive 2



Radial Cable for Drive 1

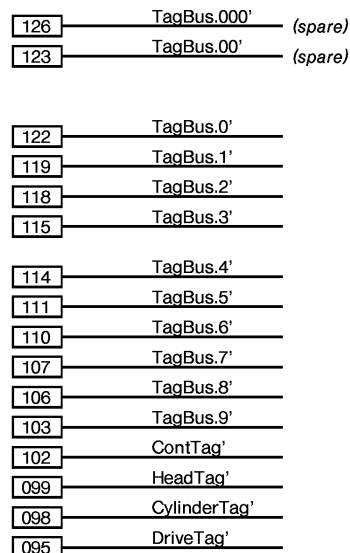


Radial Cable for Drive 3

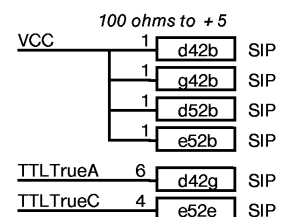
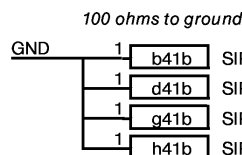
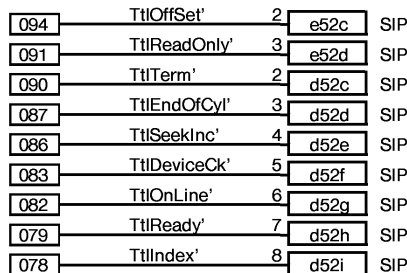


Daisy Chain Cable

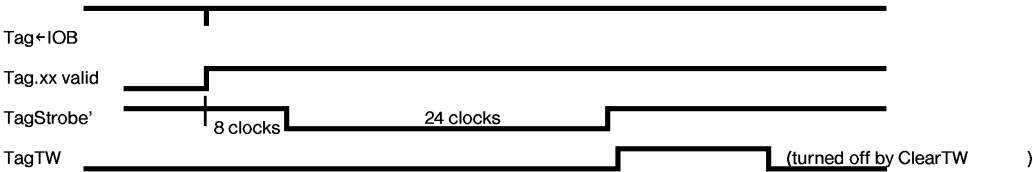
Daisy chain outputs



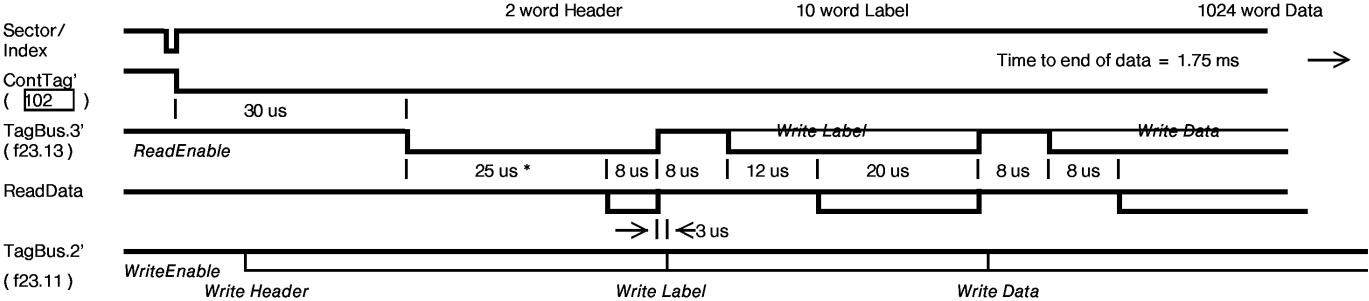
Daisy chain TTL inputs



Head or Cylinder Tag Instruction



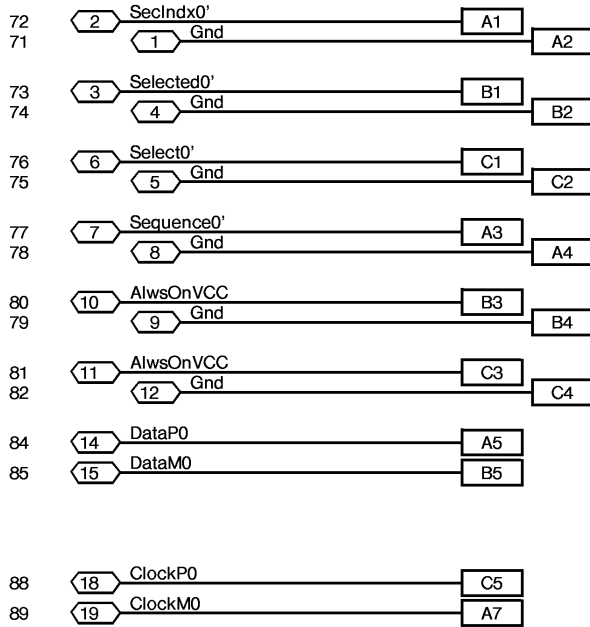
Read or Write Instruction



* This value is for reading a pack on the same drive that Headers were written
It may vary by +/- 15 us on other drives

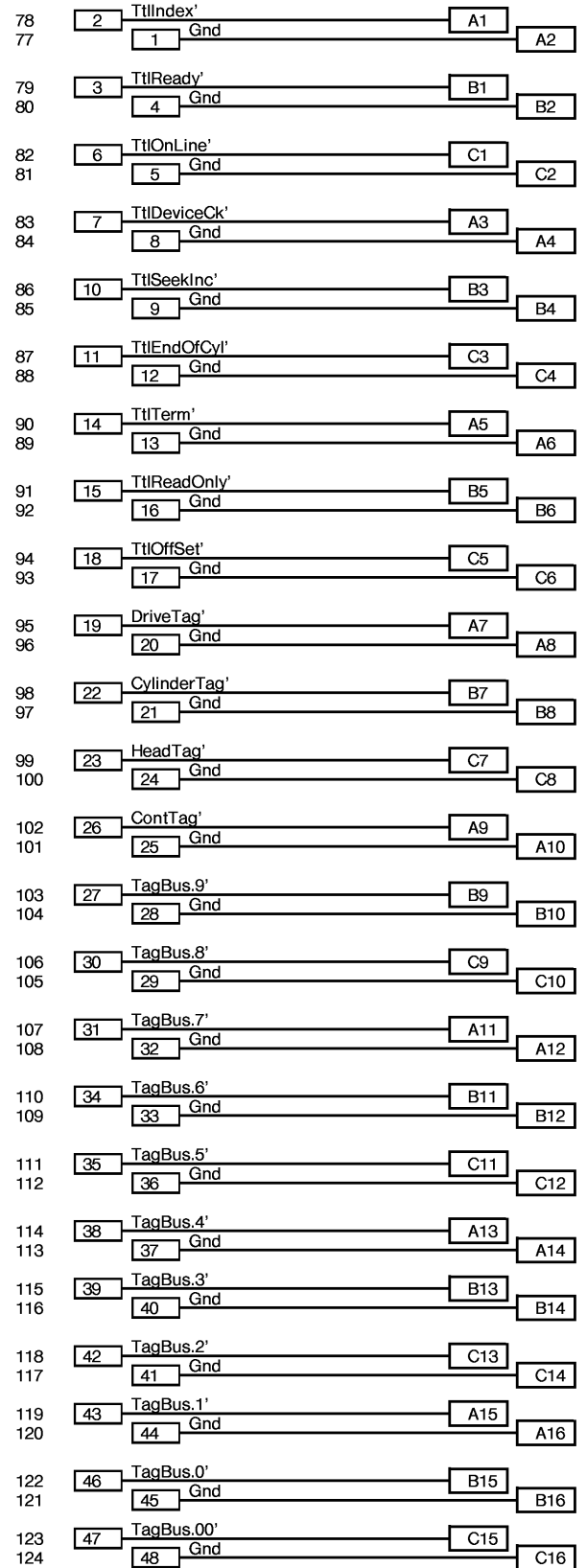
Radial Cable for Drive U

AMP 204729-1



DAISEY CHAIN CABLE

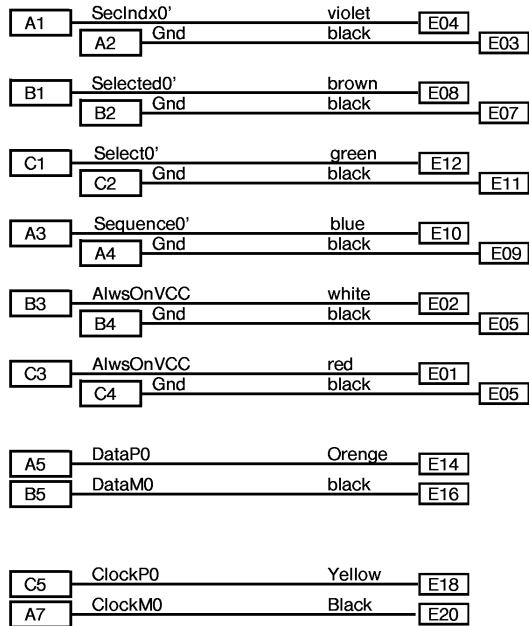
AMP 204733-1



Radial Cable for Drive U

AMP 204742-1

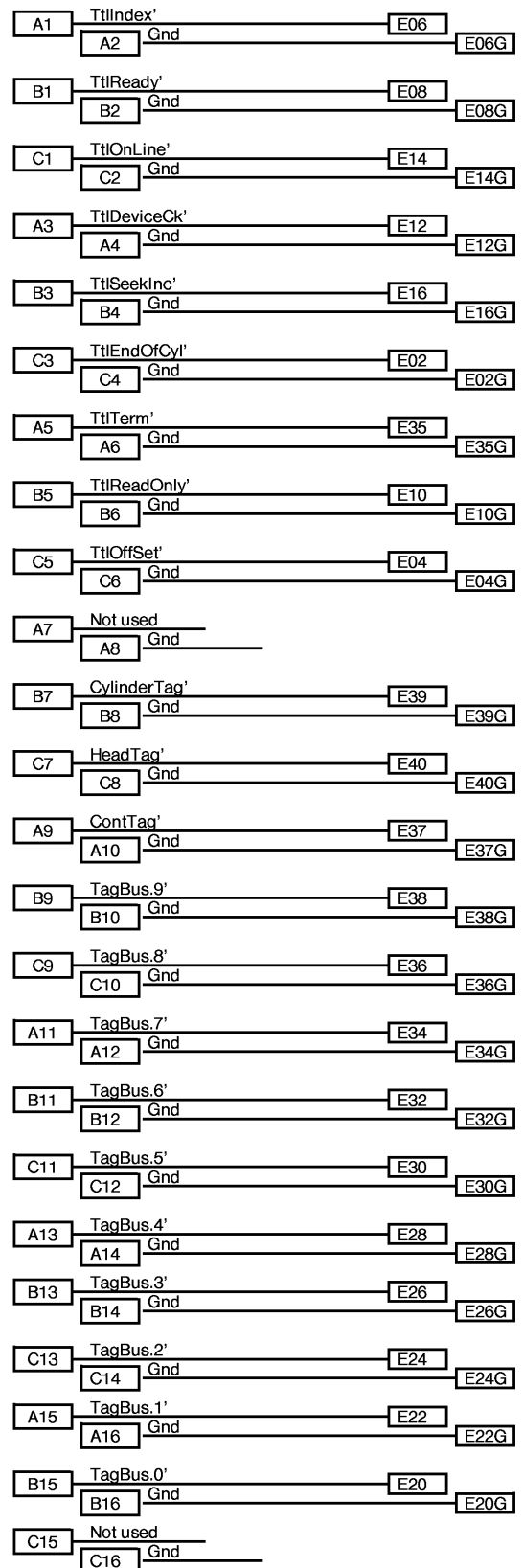
Cal-Comp
Assembly 12433

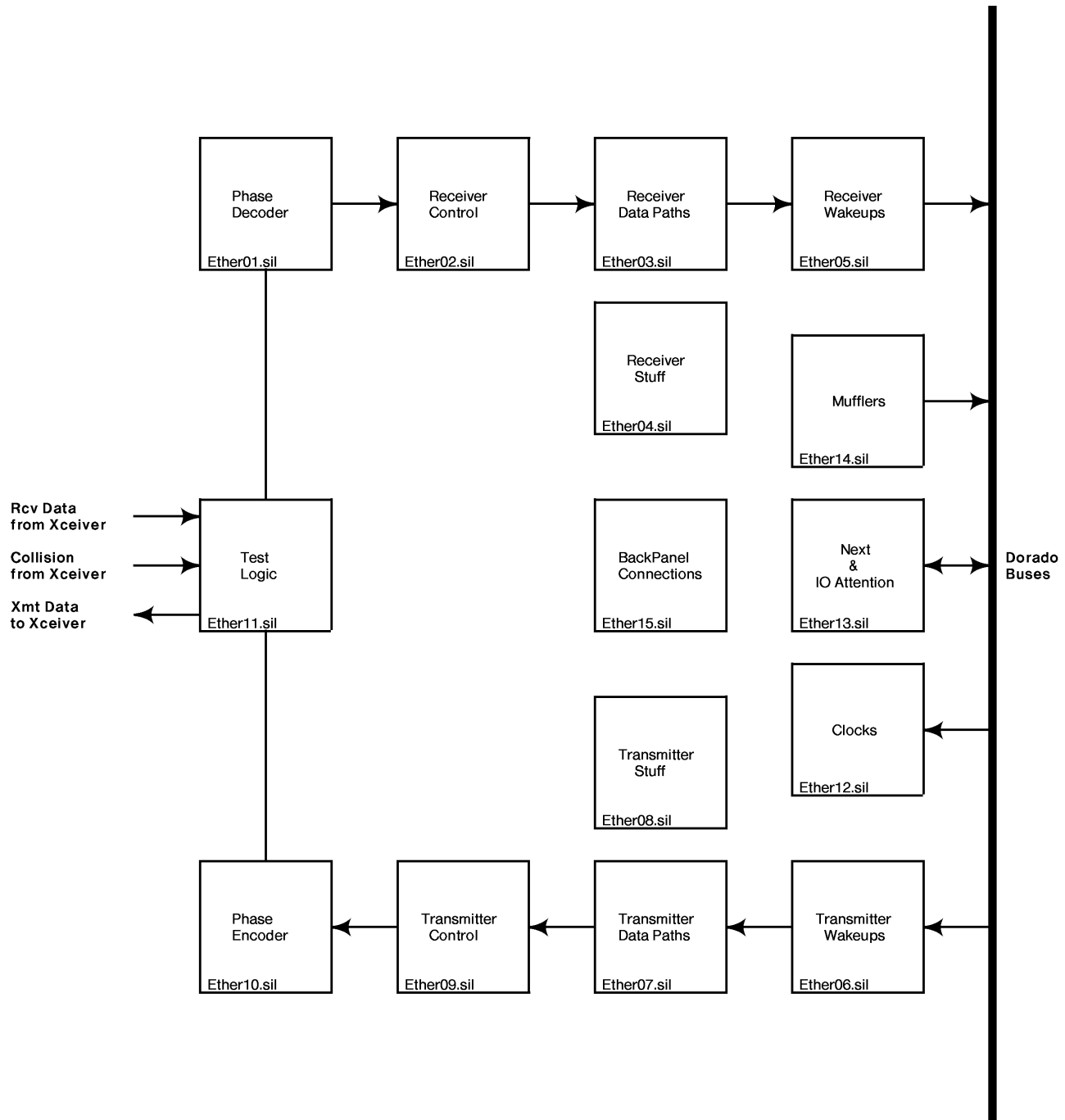


DAISEY CHAIN CABLE

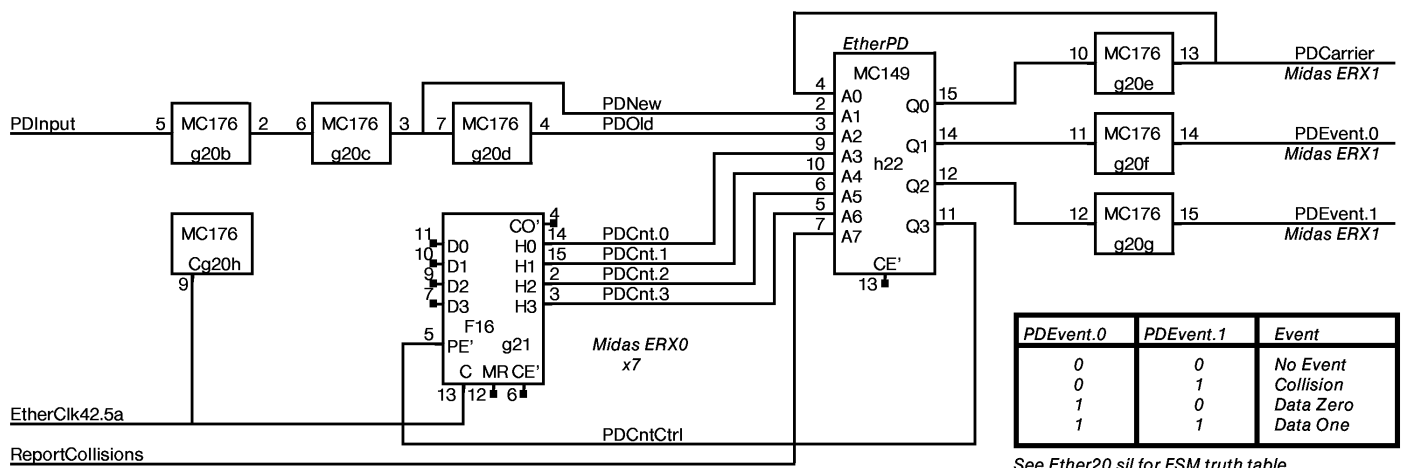
AMP 204746-1

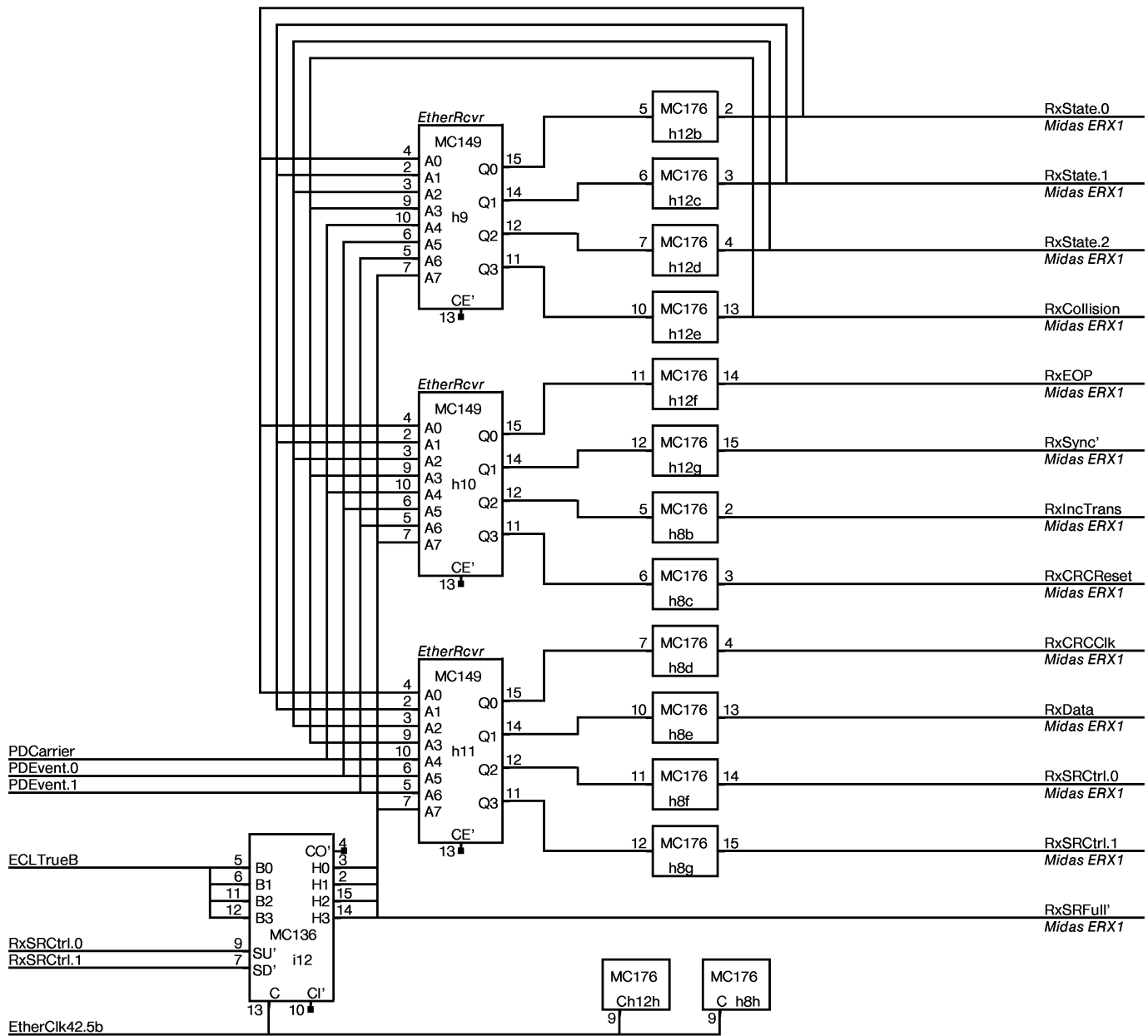
Cal-Comp
Assembly 12424





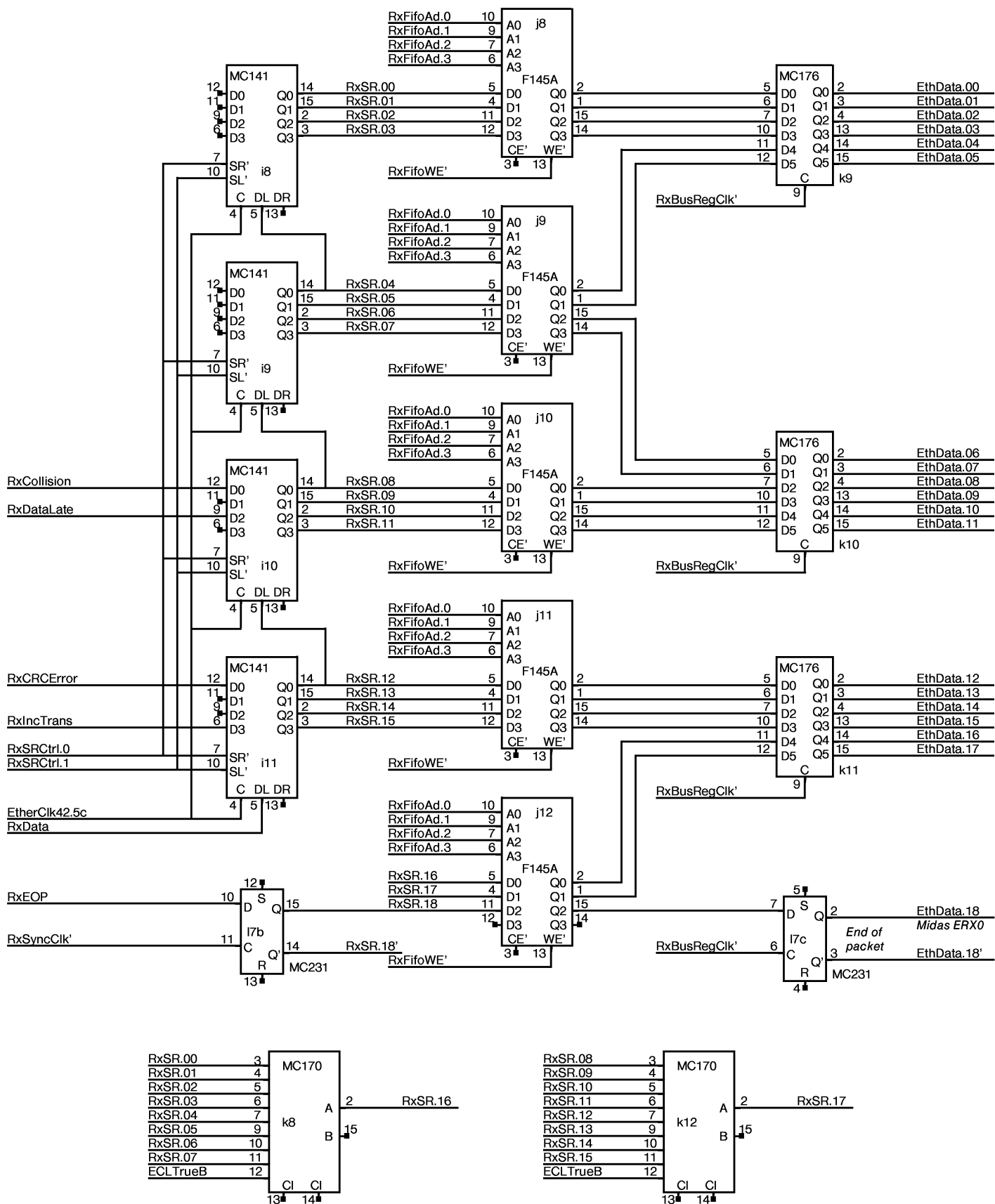
See DskEth*.sil for IOA, IOB, Muffler Control and Board Clocks

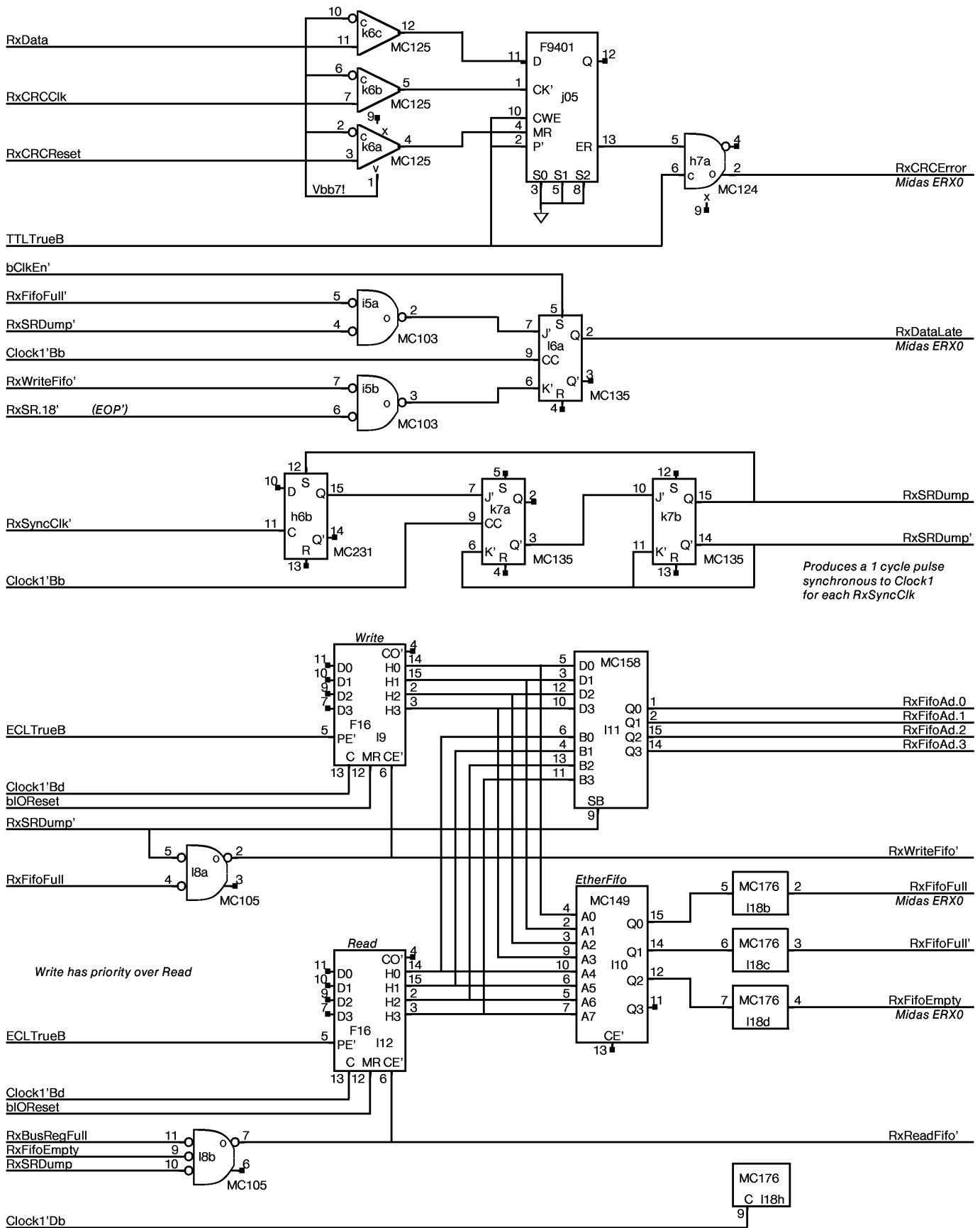


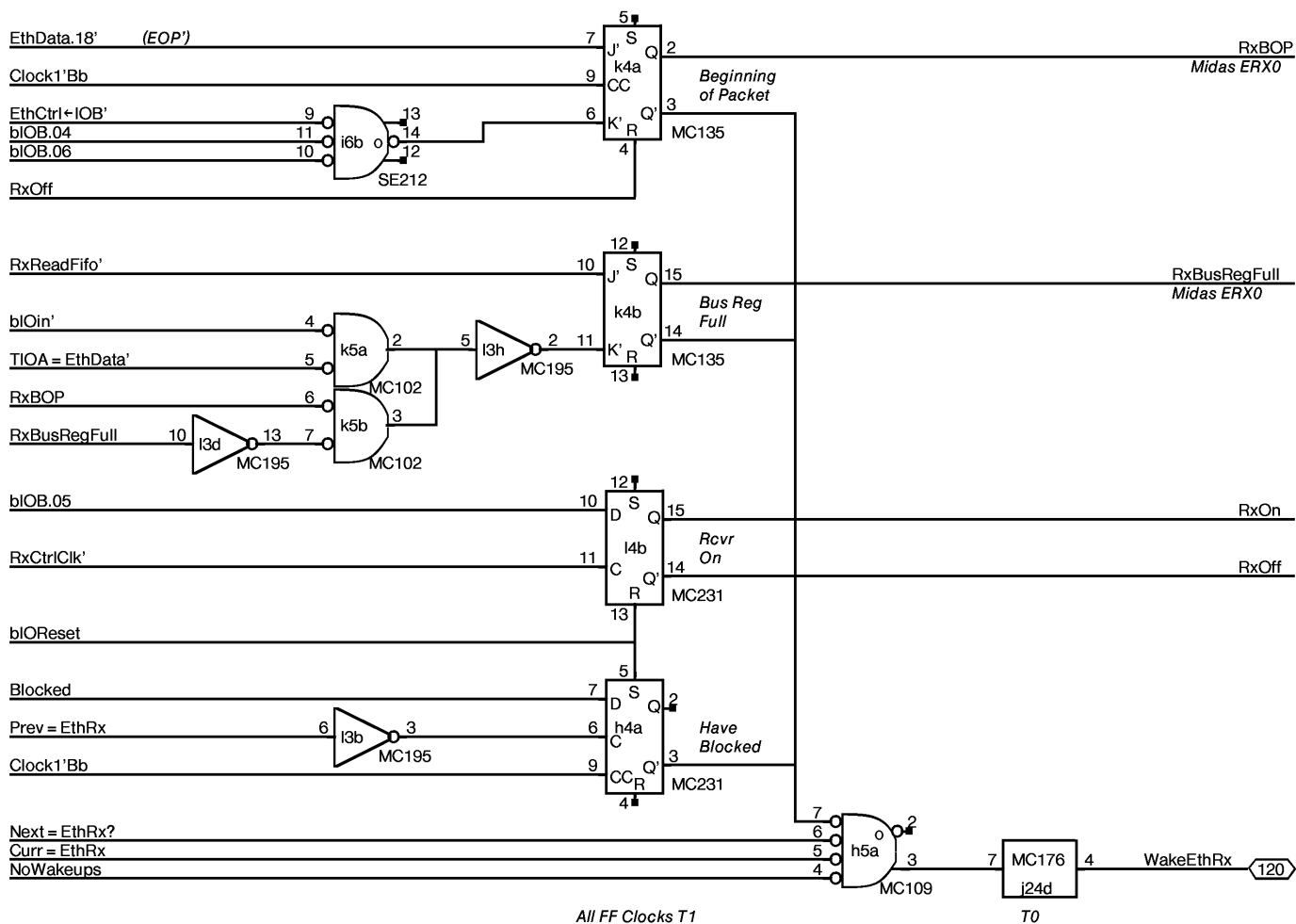


See Ether18.sil for timing diagrams

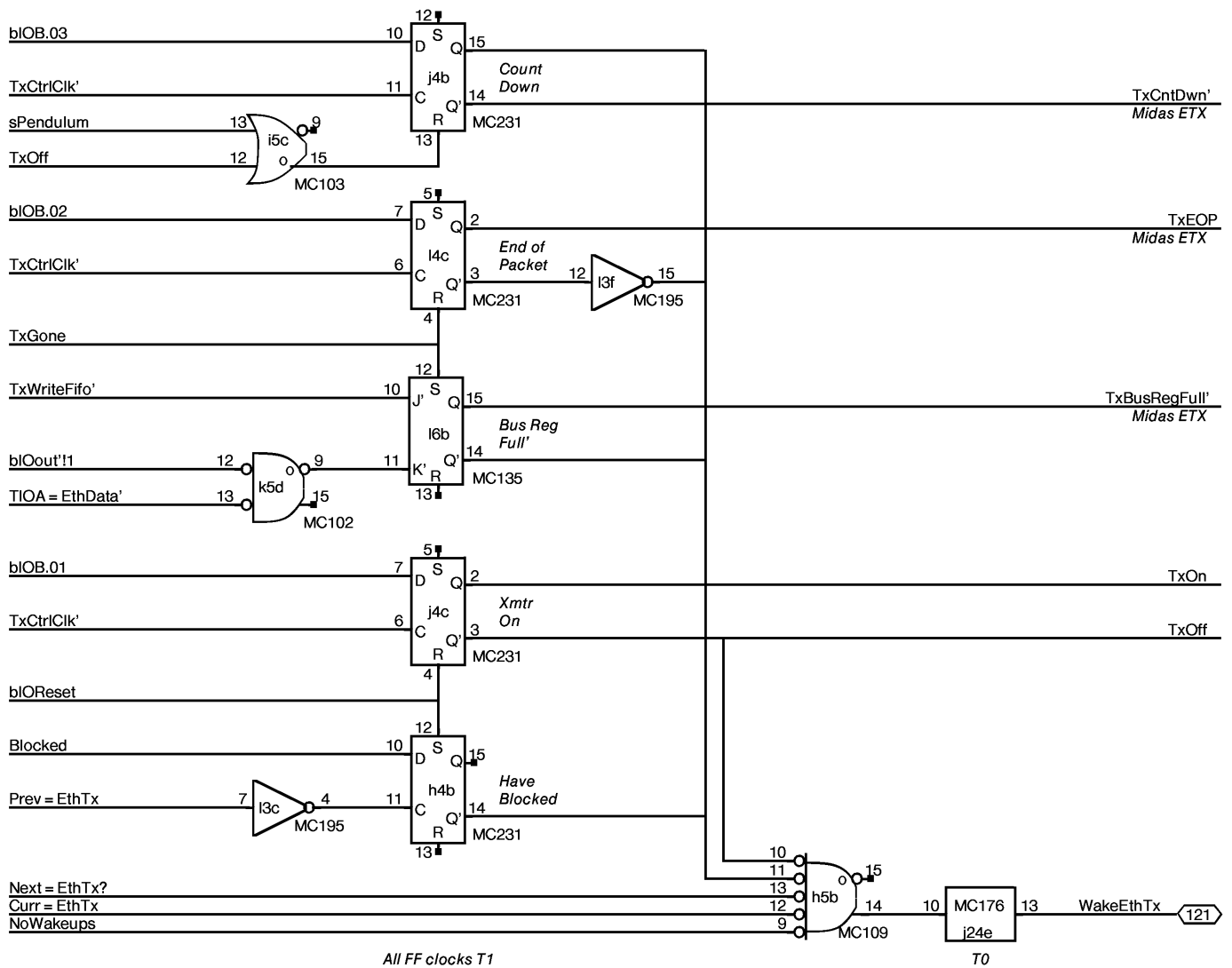
The slowest Dorado clock speed at which the receiver works is 85 ns (T0 to T1)



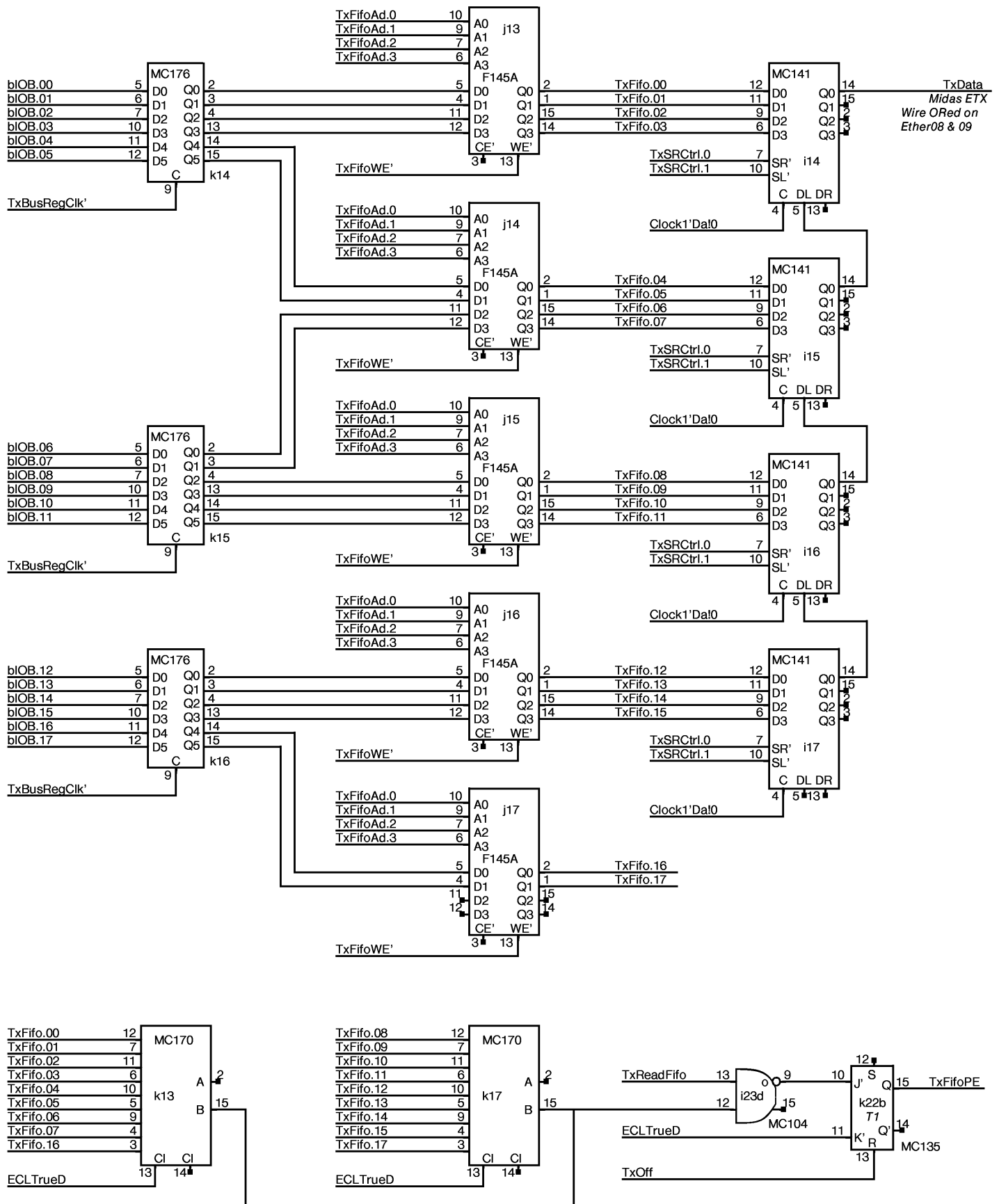


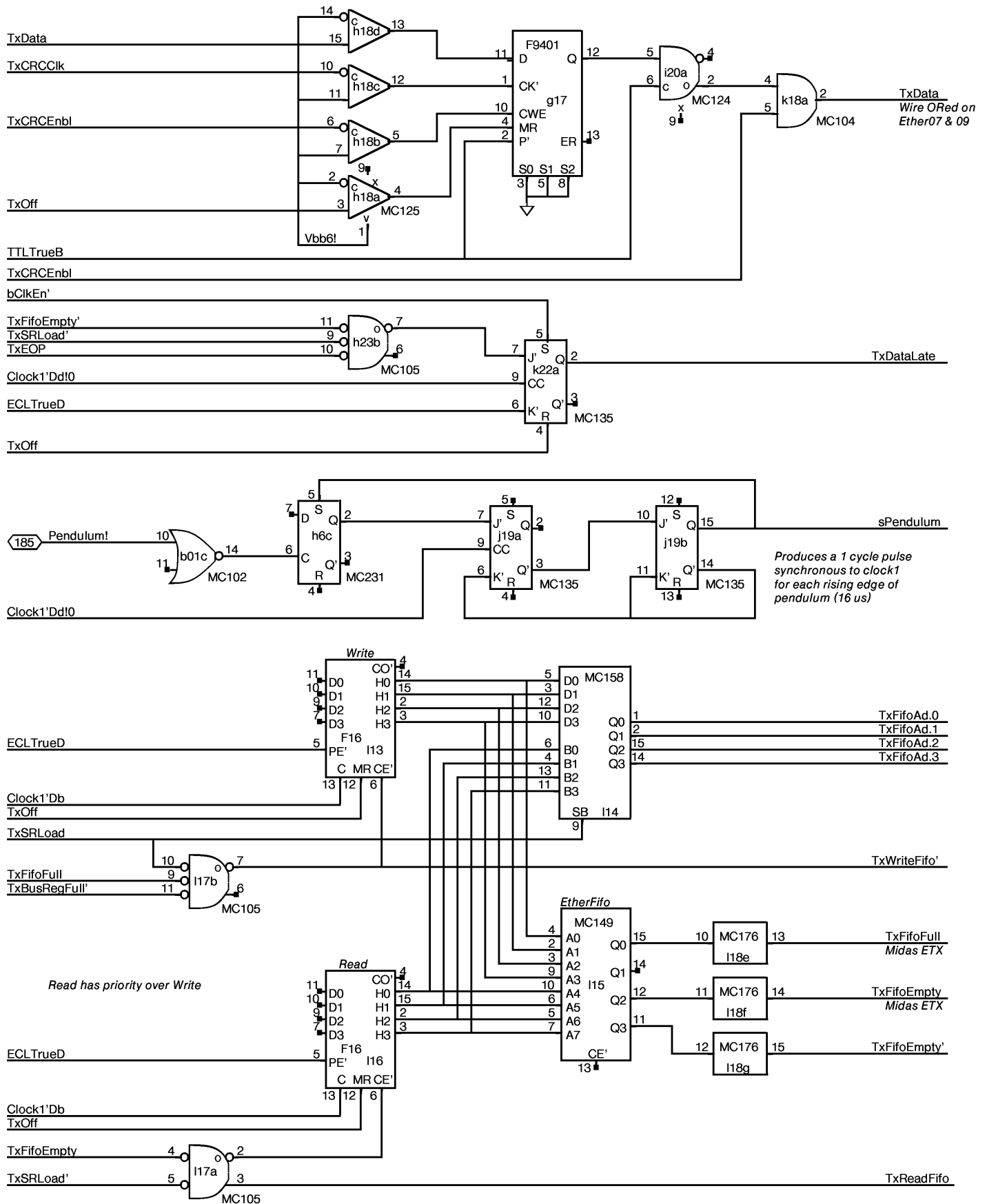


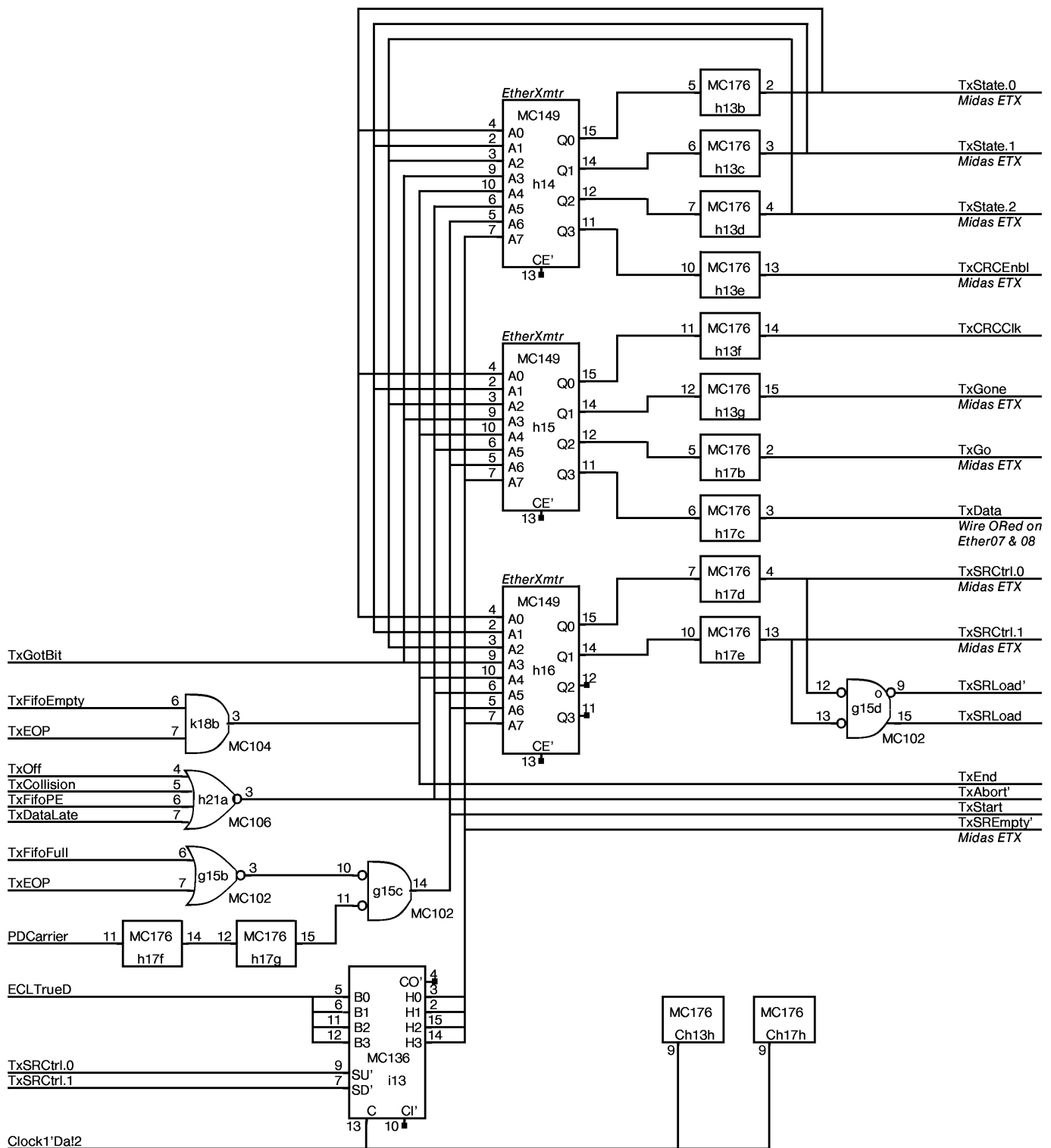
See Ether21 & 22.sil for wakeup timing diagrams



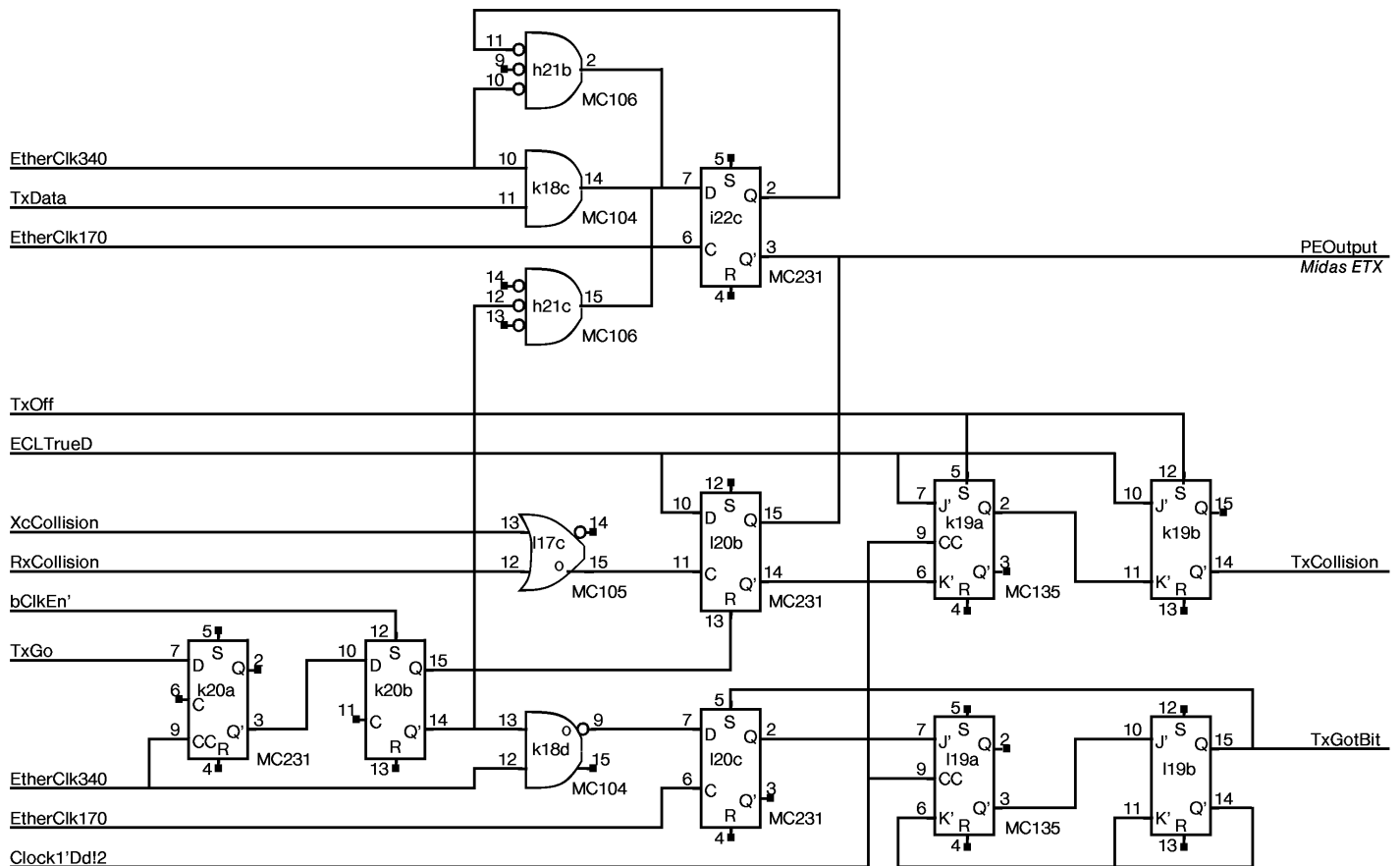
See Ether21 & 22.sil for wakeup timing diagrams







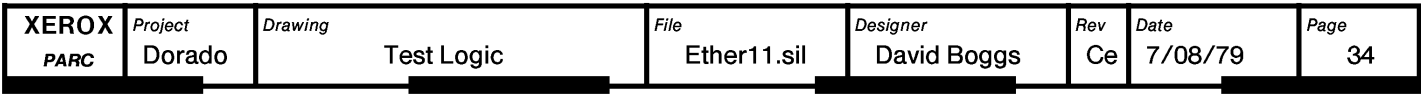
See Ether17.sil for timing diagrams

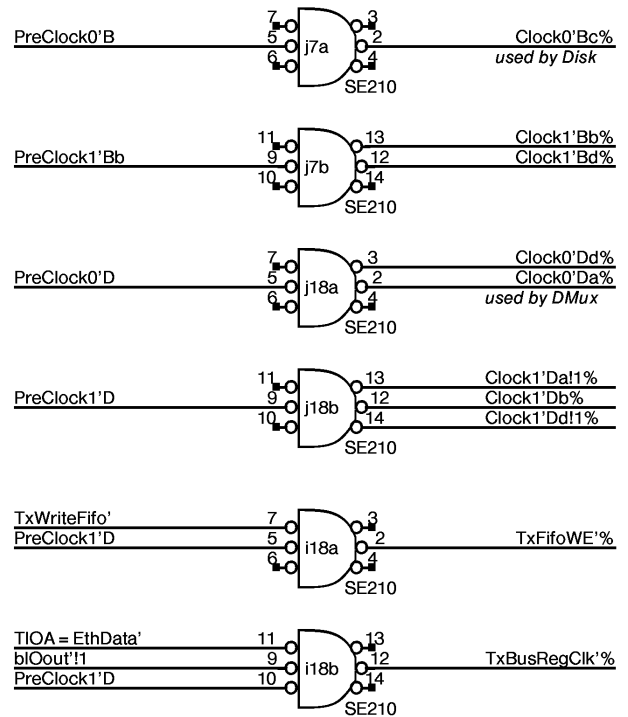
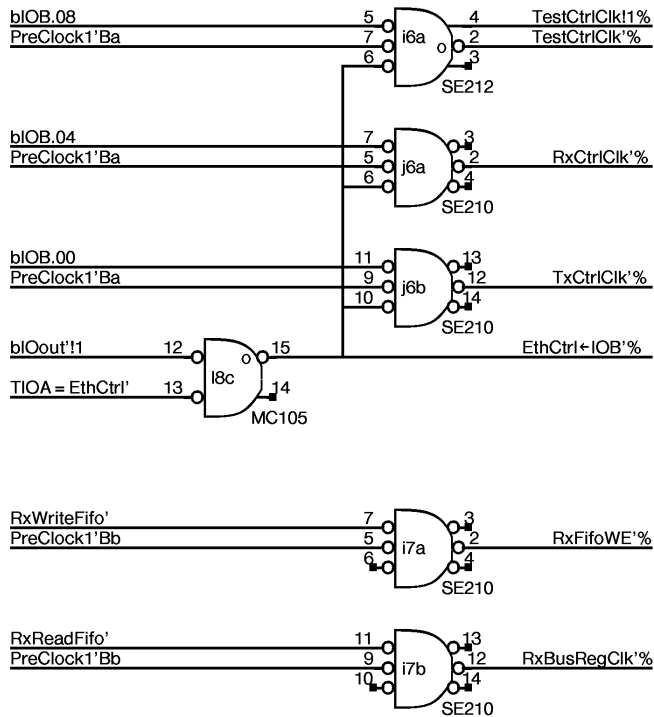


See Ether19.sil for timing diagrams

The slowest Dorado clock speed at which the transmitter works is 42.5 ns (T0 to T1)

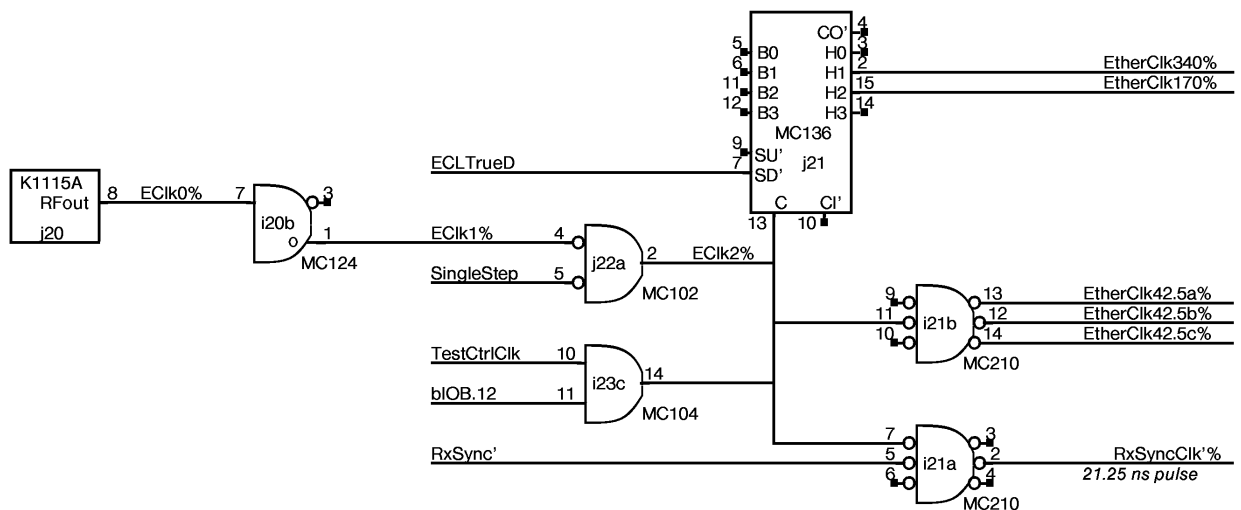
525 ns < length of jam < 915 ns

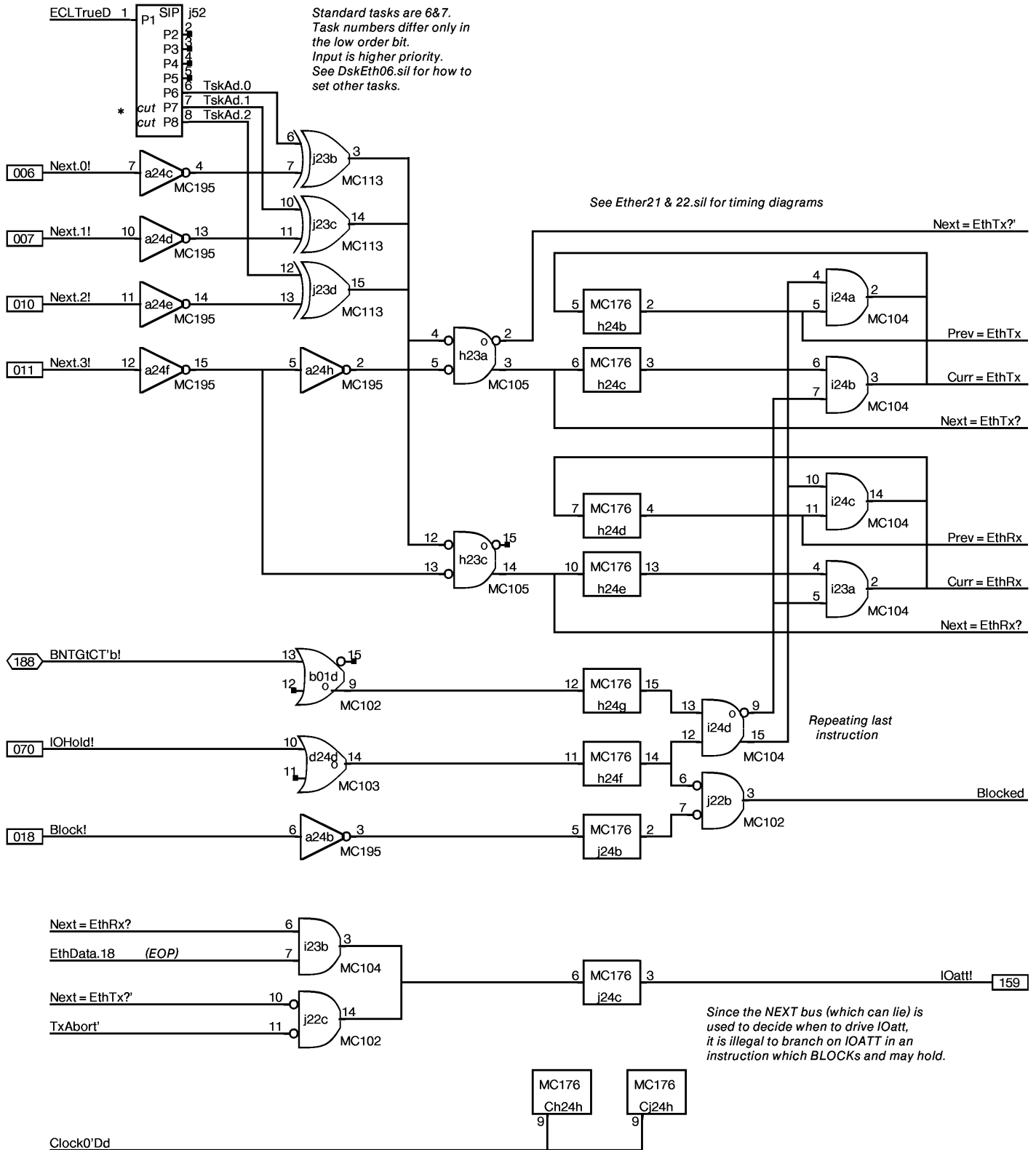


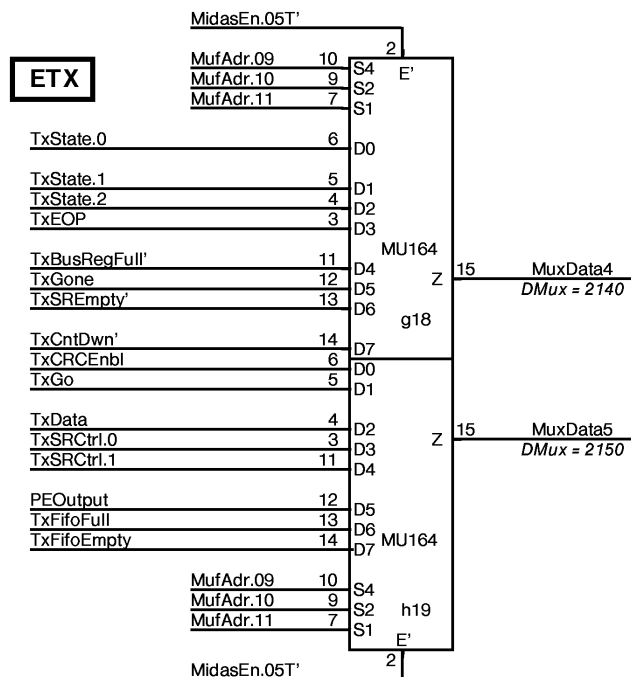
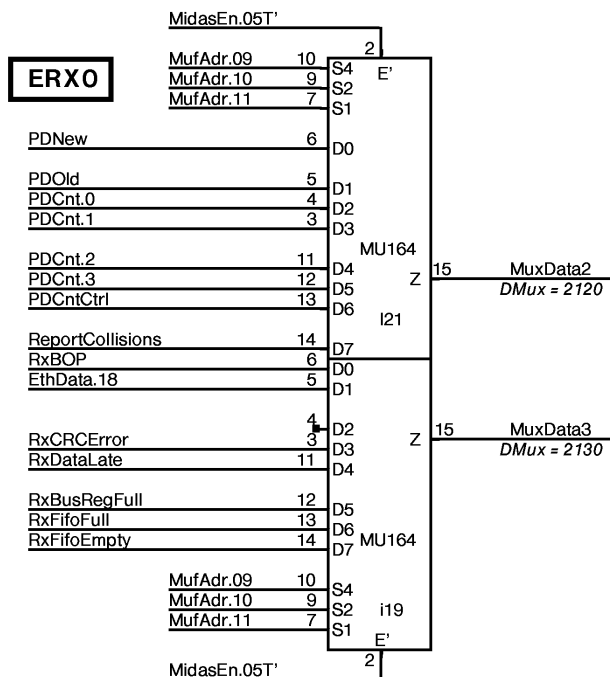


Dorado Synchronous Clocks

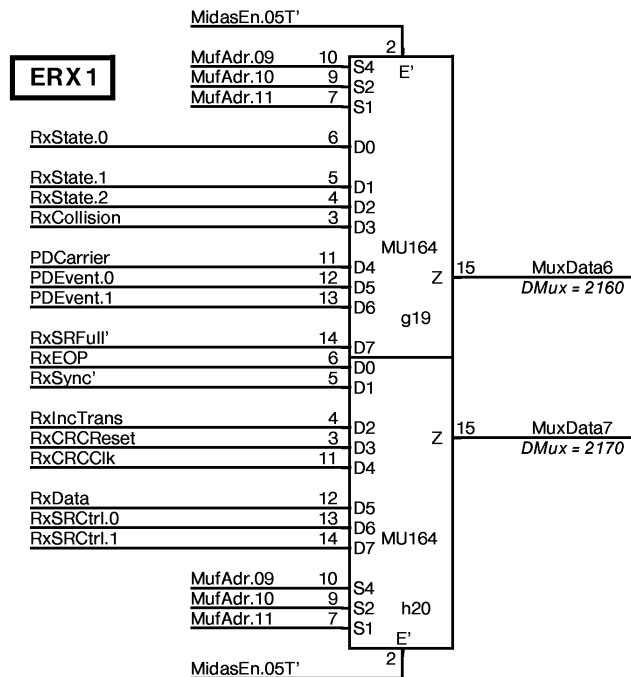
Free-running Ether Clocks





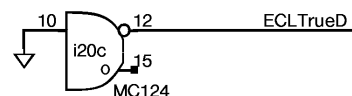
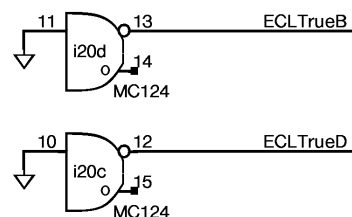
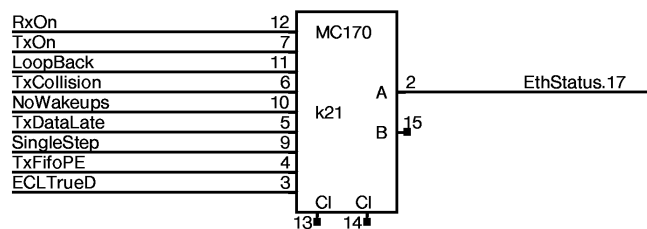
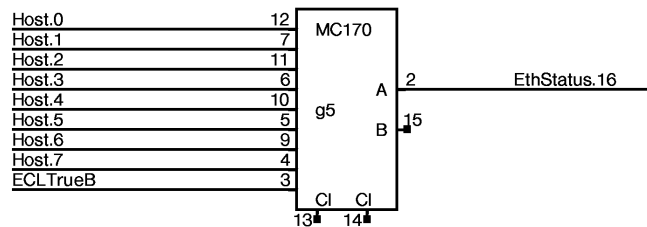
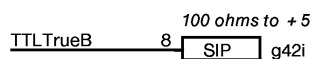
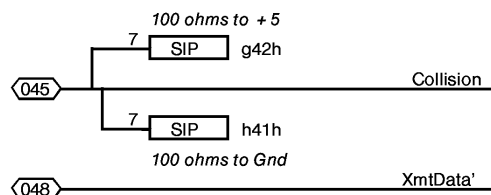
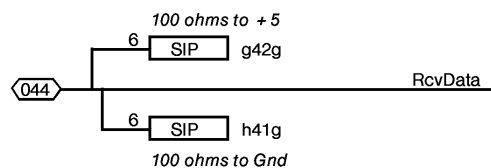


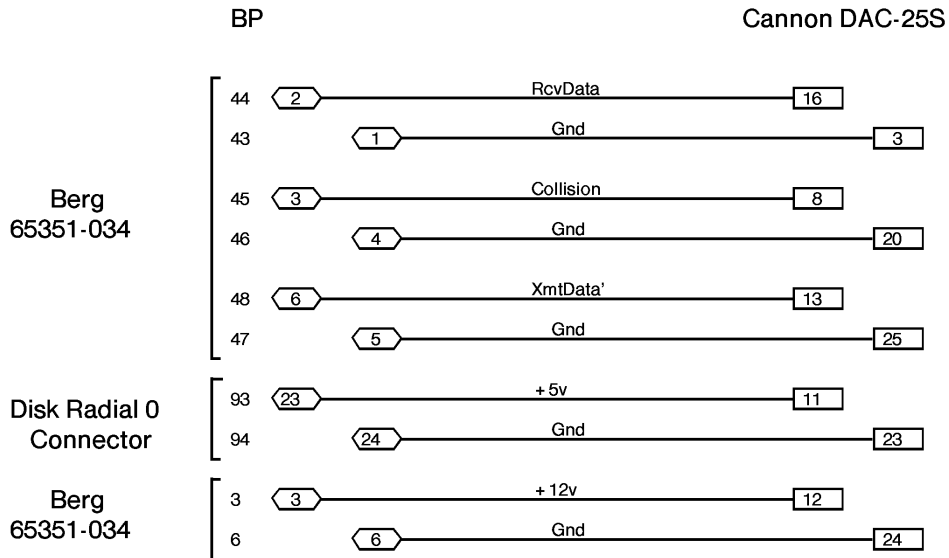
See DskEth01.sil for muffler control logic.
DMux addresses 2000-2117 are used by the disk.



To set a host address bit to 1
pull it up to gnd through 91 ohms.

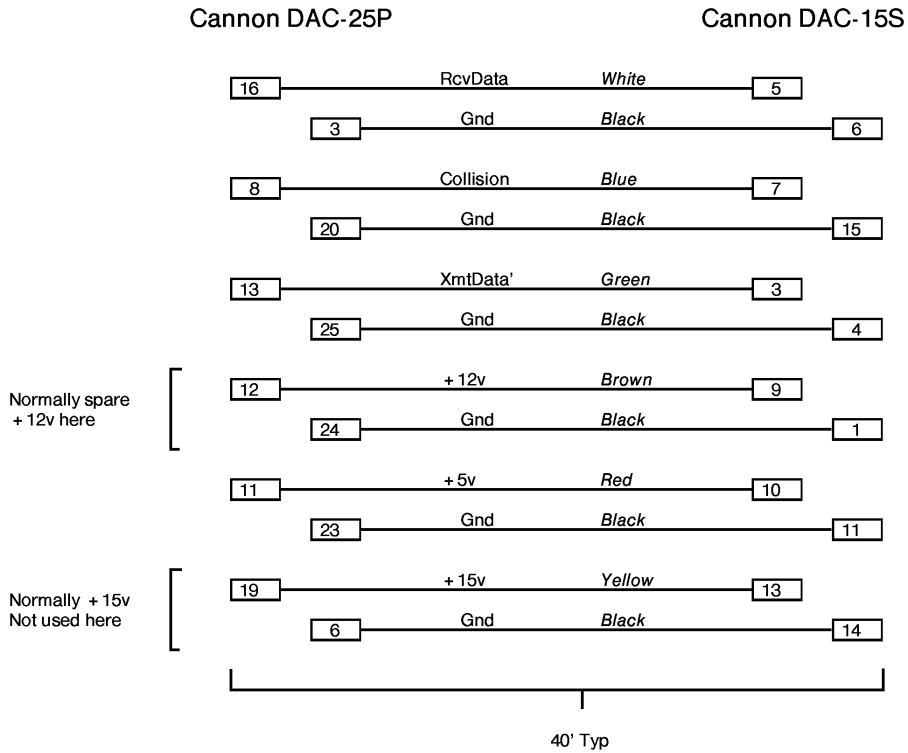
028	Host.0	200
029	Host.1	100
032	Host.2	40
033	Host.3	20
036	Host.4	10
037	Host.5	4
040	Host.6	2
041	Host.7	1



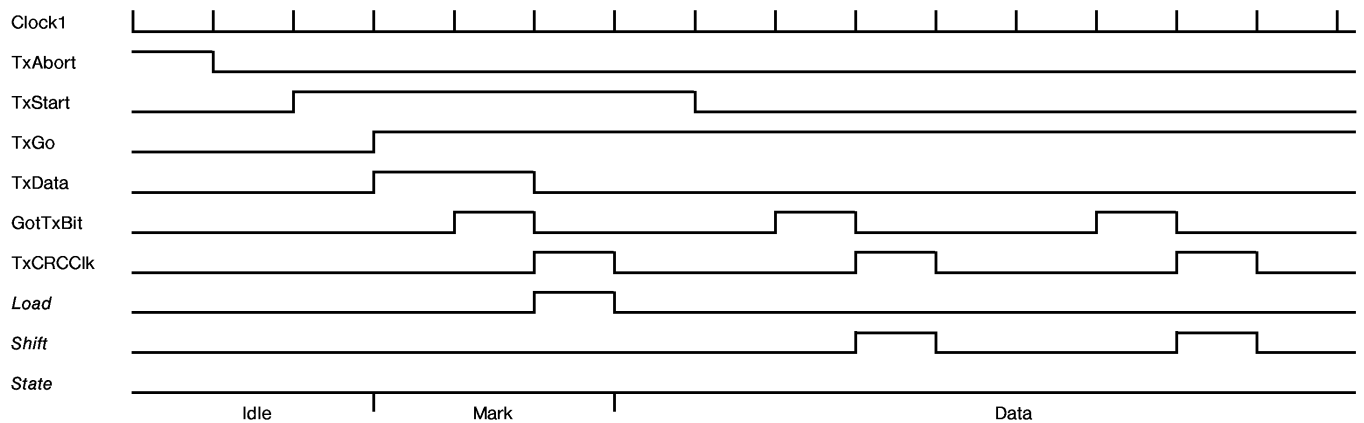


Internal Cable

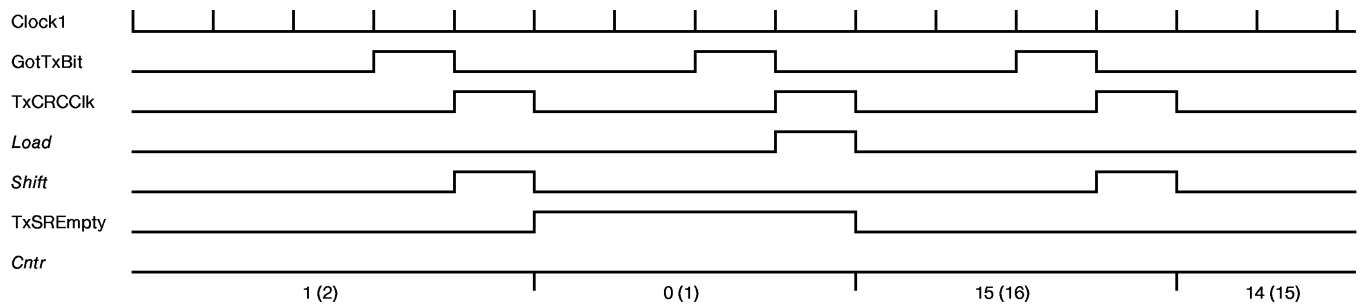
External Cable



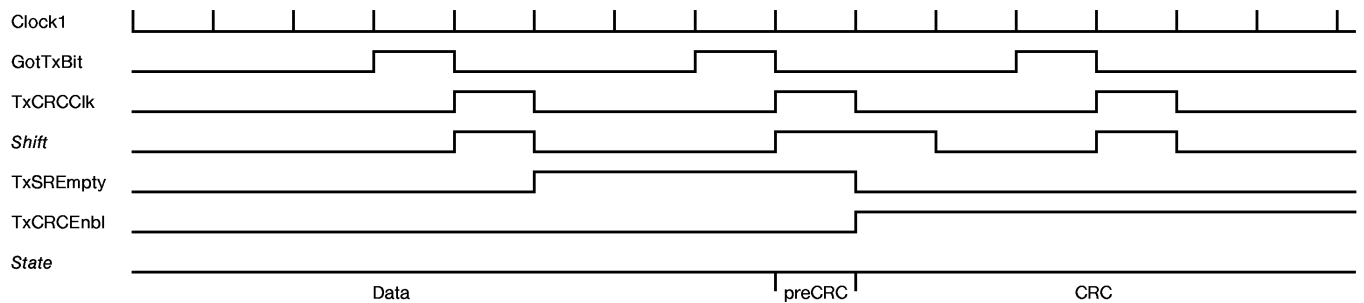
This is a standard Alto II Ethernet external cable part # 216411



Startup

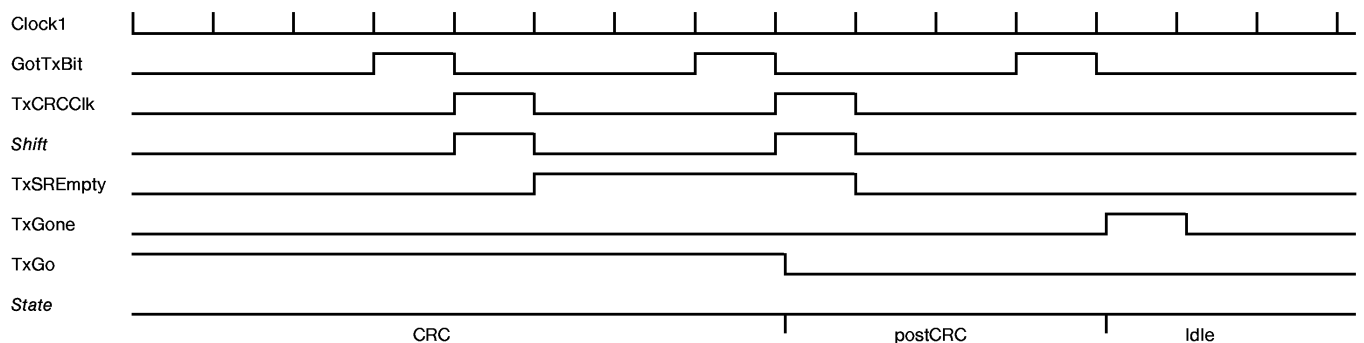


Loading Tx Shift Register

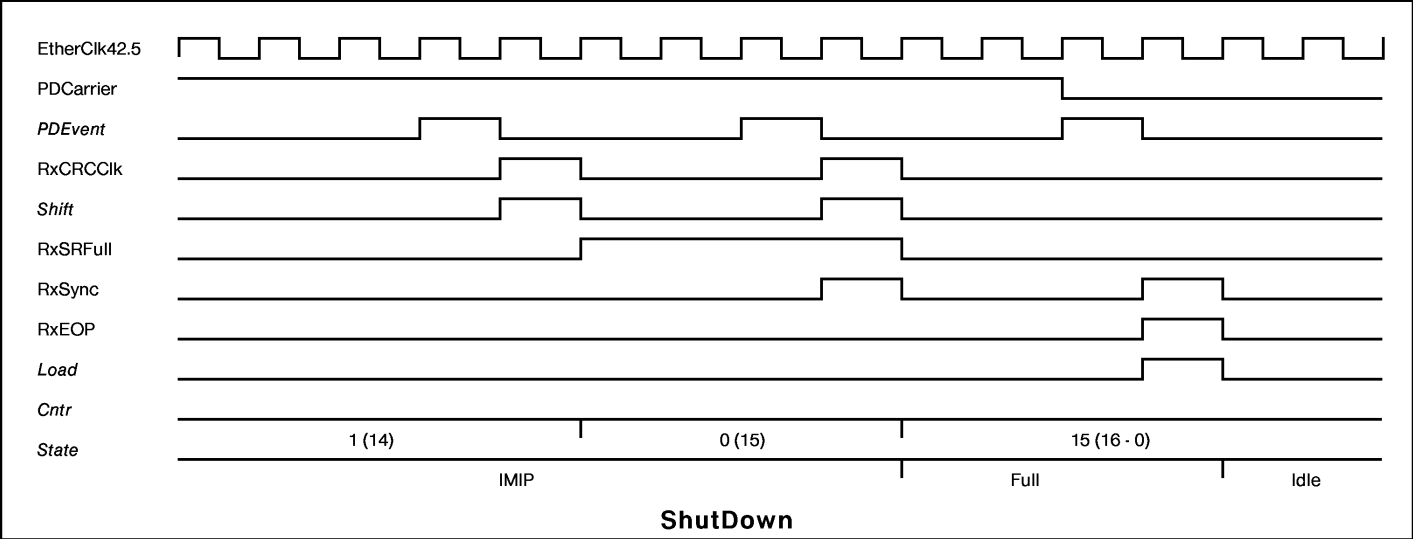
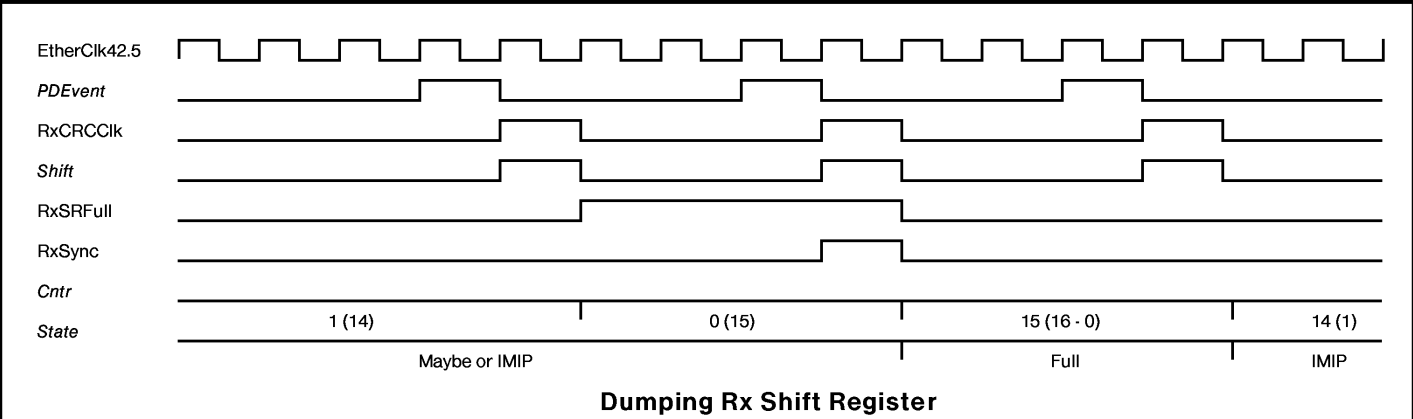
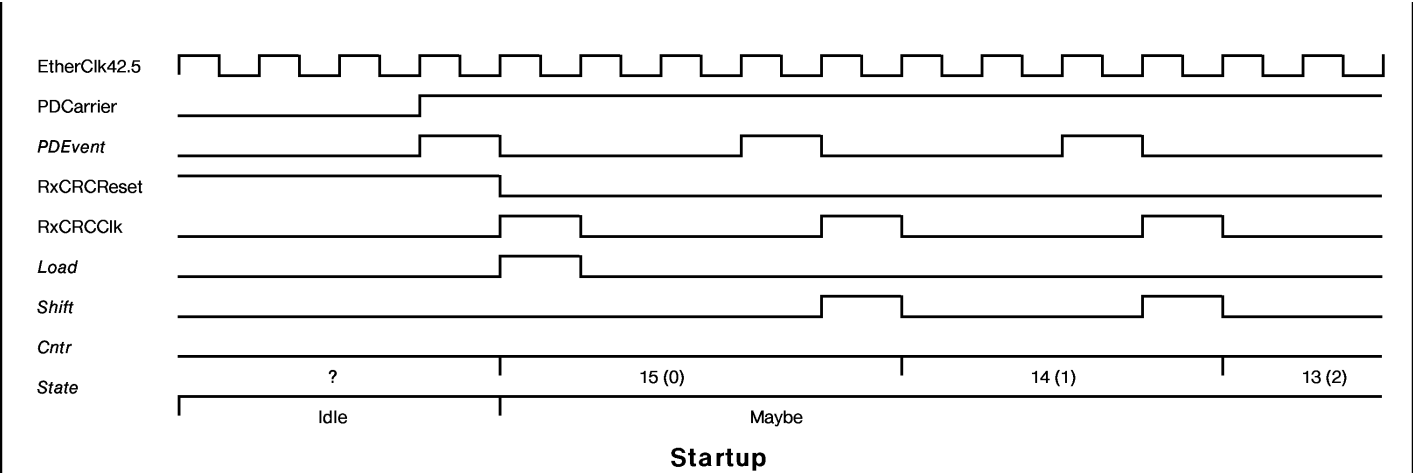


TxEnd has been high since TxSR was last loaded

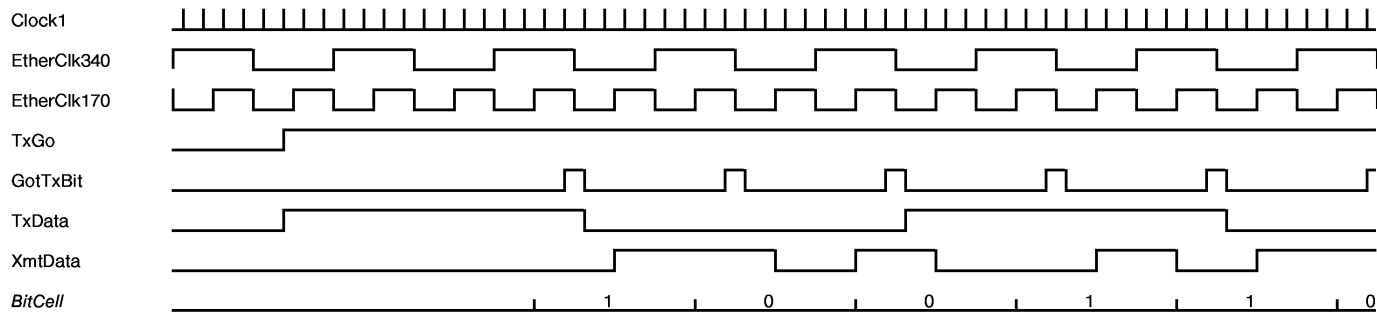
CRC



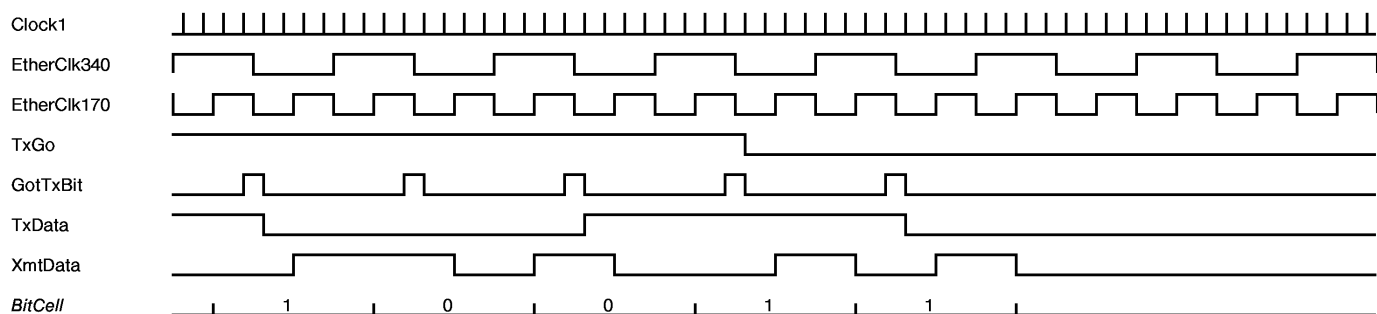
ShutDown



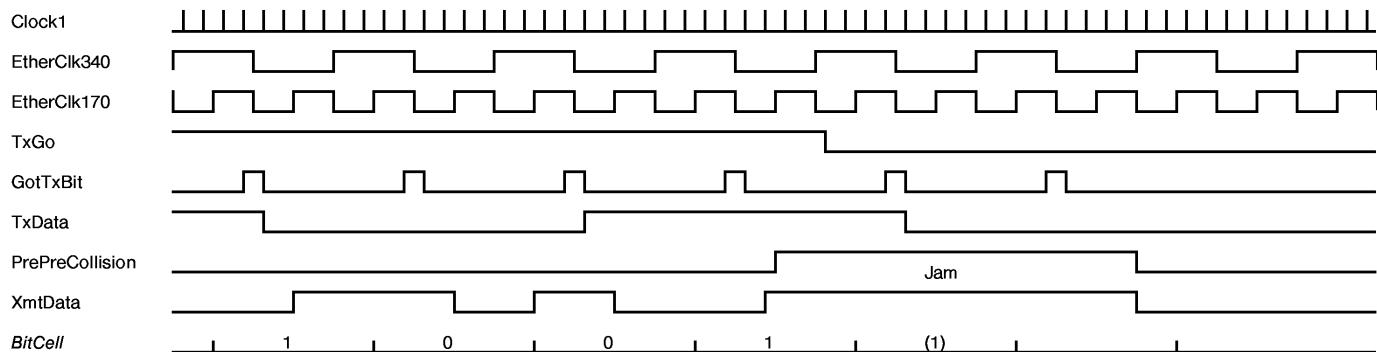
Cntr notation is <Cntr value><# occupied postions>
Phase decoder events are encoded in PDEvent.x
Shift and Load are encoded in RxSRCtrl.x
state is encoded in RxState.x
shift also decrements Cntr



Startup

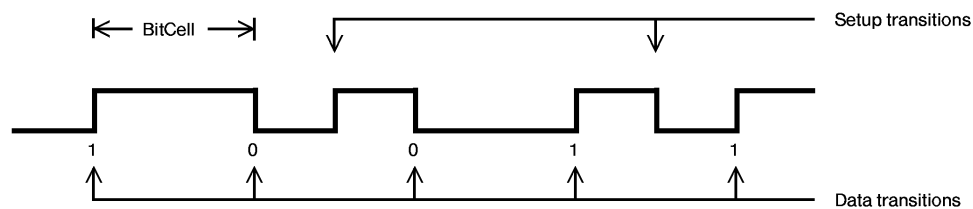


Shutdown

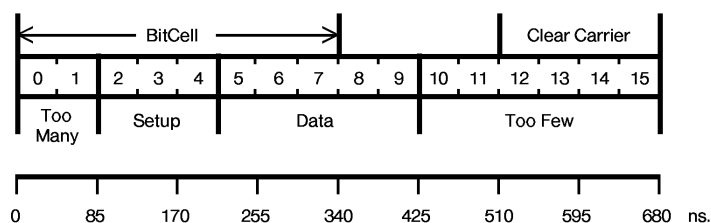


PrePreCollision is the output of the first stage of the Collision synchronizer

Collision

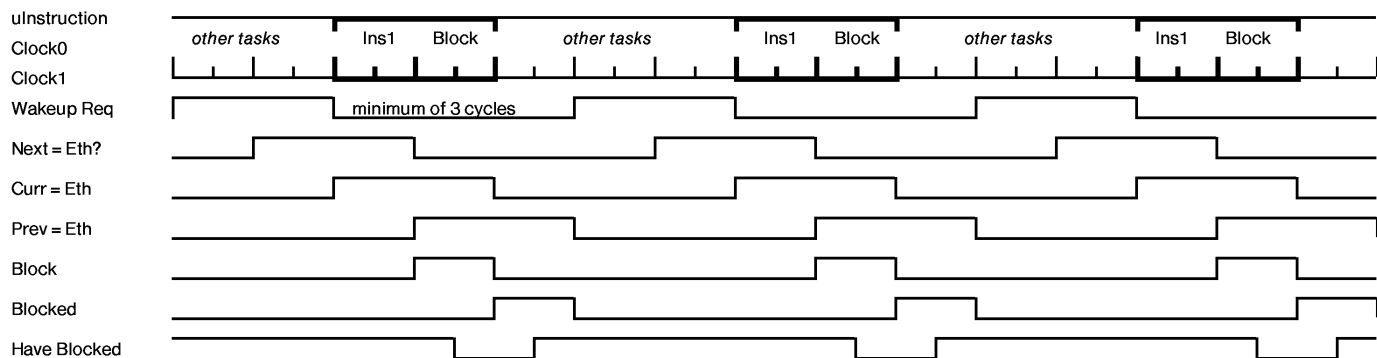


A bitcell is nominally 340 ns.
A sample is nominally 42.5 ns.

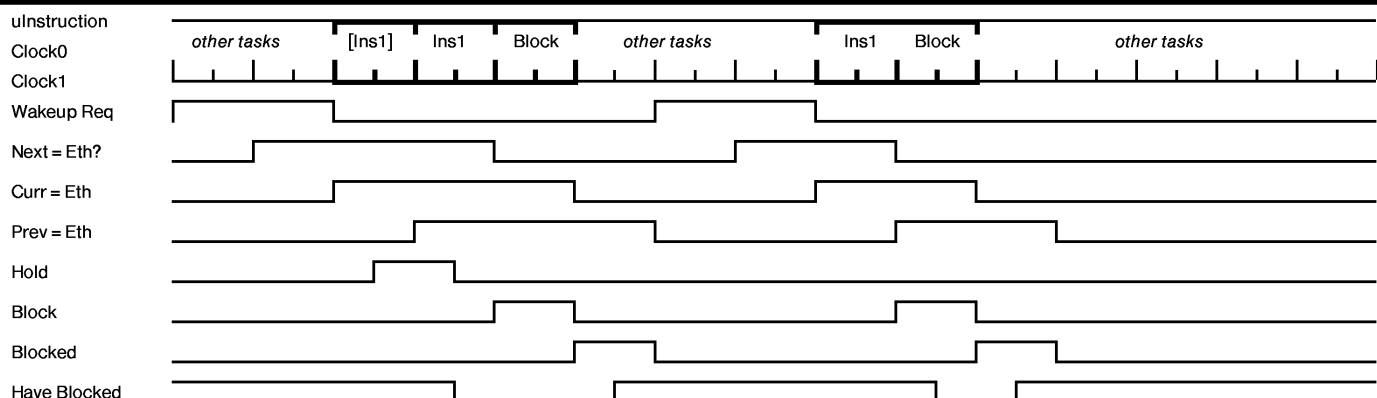


Inputs				Outputs			Comment
Carrier	Old	New	Cnt	Carrier	Event	CntCtrl	
Low	Low	Low	d/c	Low	NoEvent	Count	Idle
Low	Low	High	d/c	High	One	Reset	Start of packet (start bit)
Low	High	Low	d/c	High	Collision	Reset	Impossible
Low	High	High	d/c	Low	NoEvent	Count	Impossible
High	Low	High	0-1	High	Collision	Count	Too many transitions
High	High	Low	0-1	High	Collision	Count	Too many transitions
High	Low	High	2-4	High	NoEvent	Count	Setup transition (zero next)
High	High	Low	2-4	High	NoEvent	Count	Setup transition (one next)
High	Low	High	5-9	High	One	Reset	Data transition
High	High	Low	5-9	High	Zero	Reset	Data transition
High	Low	High	10-15	High	Collision	Reset	Too few transitions
High	High	Low	10-15	High	Collision	Reset	Too few transitions
High	Low	Low	0-11	High	NoEvent	Count	Active
High	High	High	0-11	High	NoEvent	Count	Active
High	Low	Low	12-15	Low	Collision	Reset	End of packet
High	High	High	12-15	High	Collision	Reset	Jam

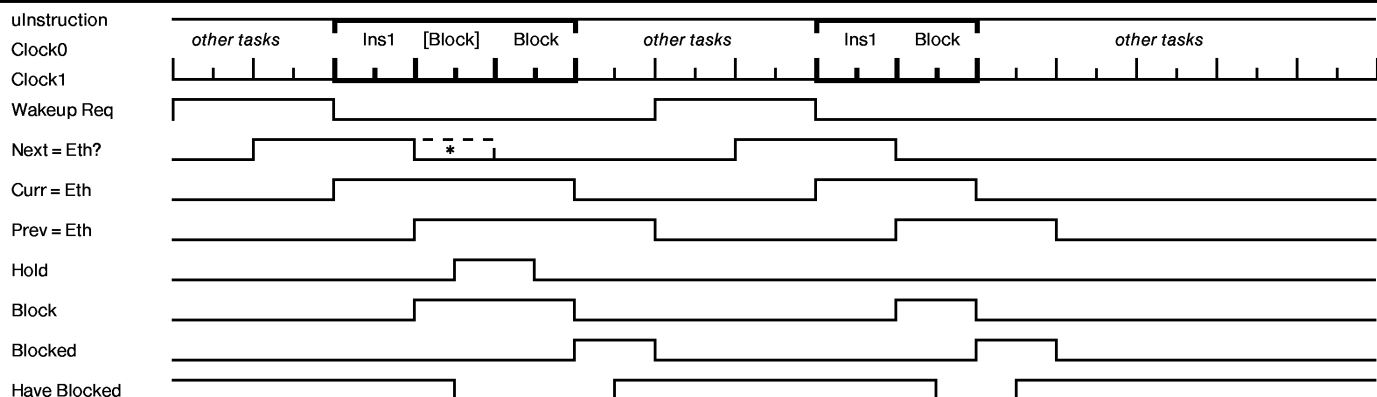
"Impossible" conditions can happen right after power up.
d/c means "don't care".



Normal case

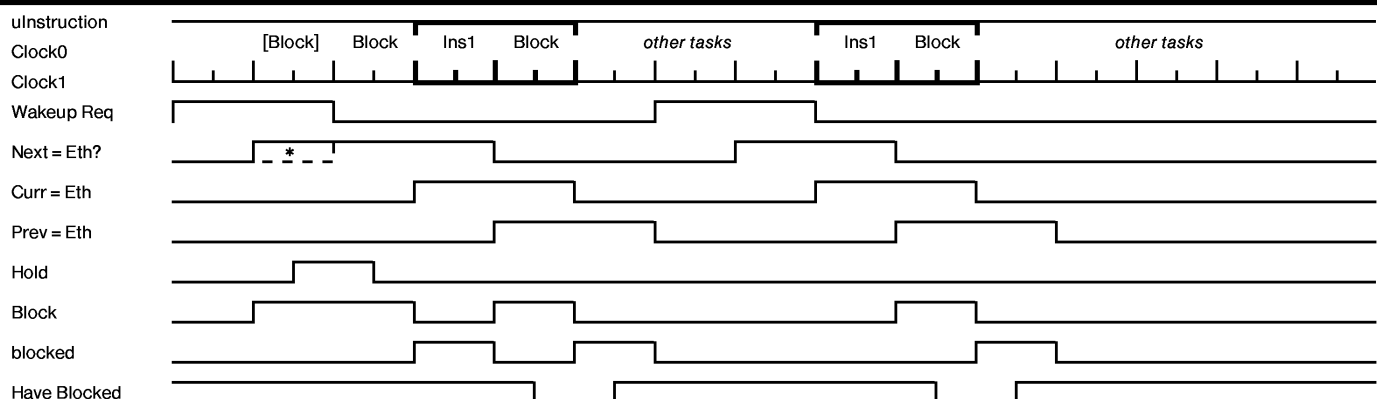


Hold in first instruction & not preempted



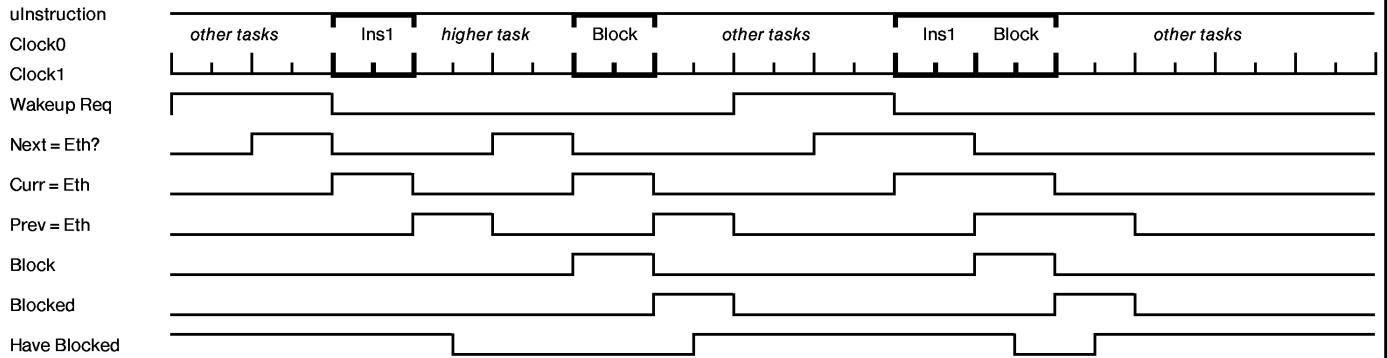
* NEXT bus is wrong

Block & Hold & not preempted

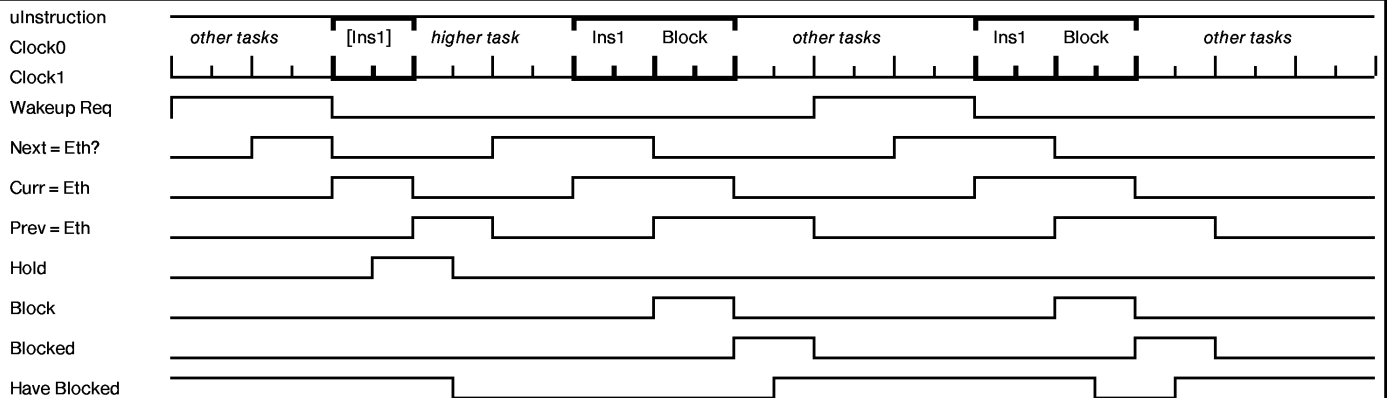


* NEXT bus is wrong

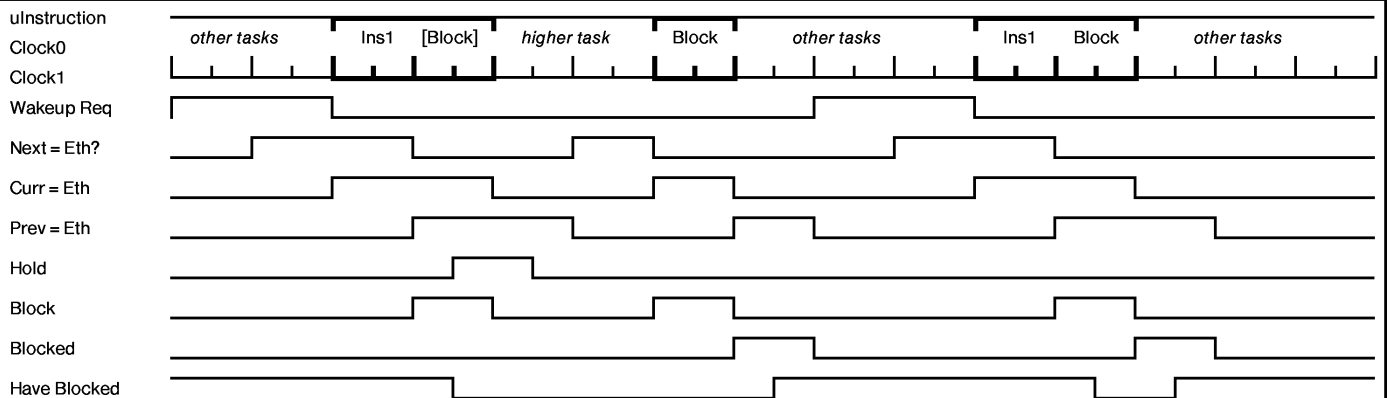
Block & Hold & (BNT = Eth < CTask)



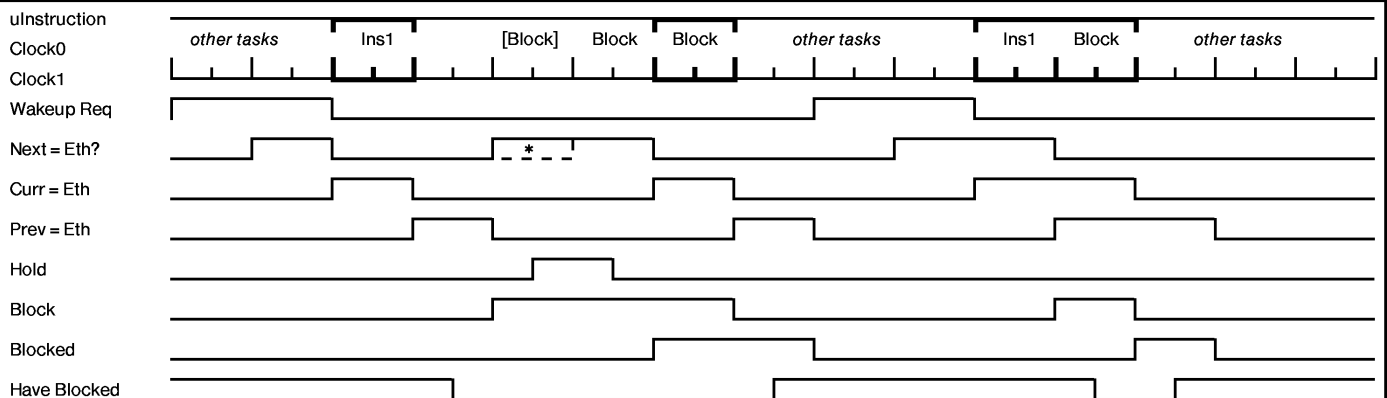
Preemption after first instruction



Hold in first instruction & then preempted



Block & Hold & then preempted



* NEXT bus is wrong

Block & Hold & (BNT = Eth < CTask) while Eth is preempted

00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
Tx Cmd Enbl'	Tx On	Tx EOP	Tx Cnt Dwn	Rx Cmd Enbl'	Rx On	Rx BOP'		Test Cmd Enbl'	Loop Back	Single Step	No Wake ups	Test Clock	Test Coll'	Test Data	Report Colls

Output

00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
Host Address								Rx On	Tx On	Loop Back	Tx Coll	No Wake ups	Tx Data Late	Single Step	Tx Fifo PE
028	029	032	033	036	037	040	041								

Input

The host address is set by jumpers on the right backplane.
To set a bit to one, pull it up to ground through 91 ohms.